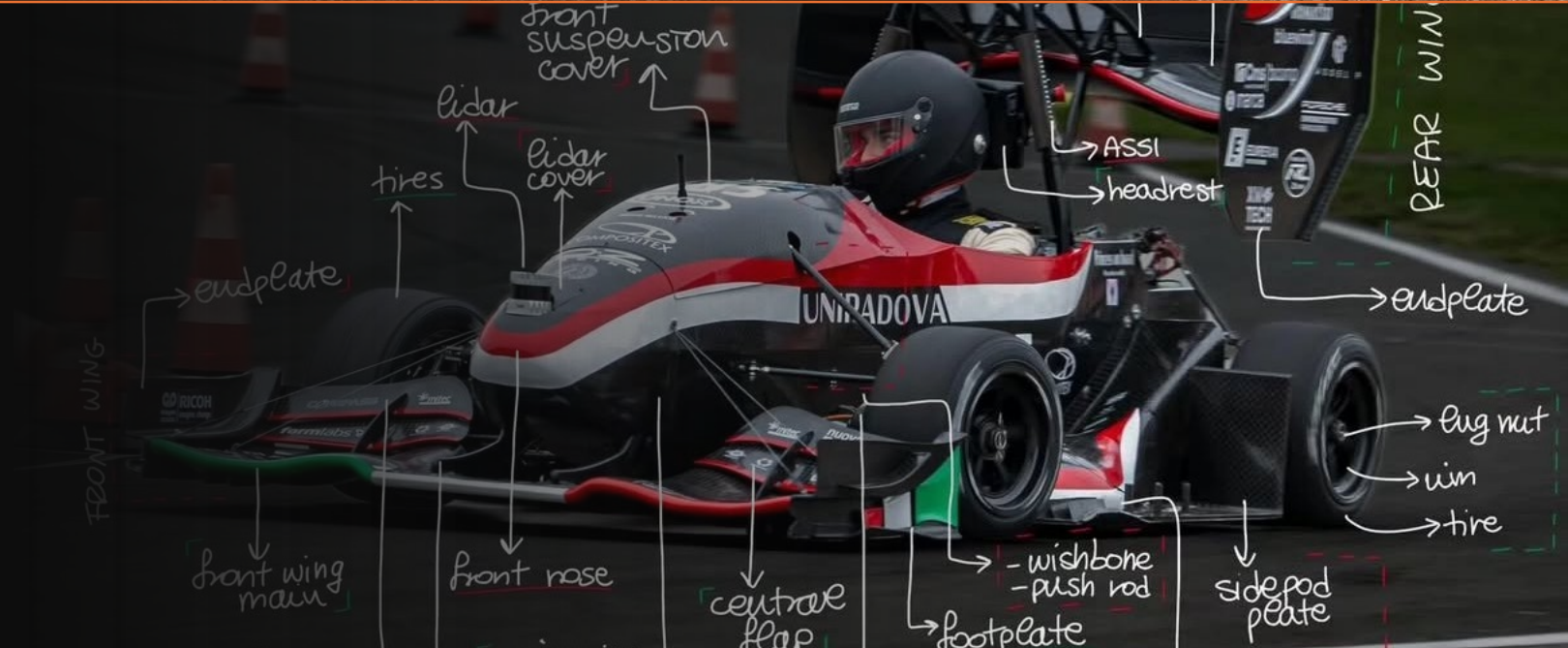


Formula Electric Driverless Presentation

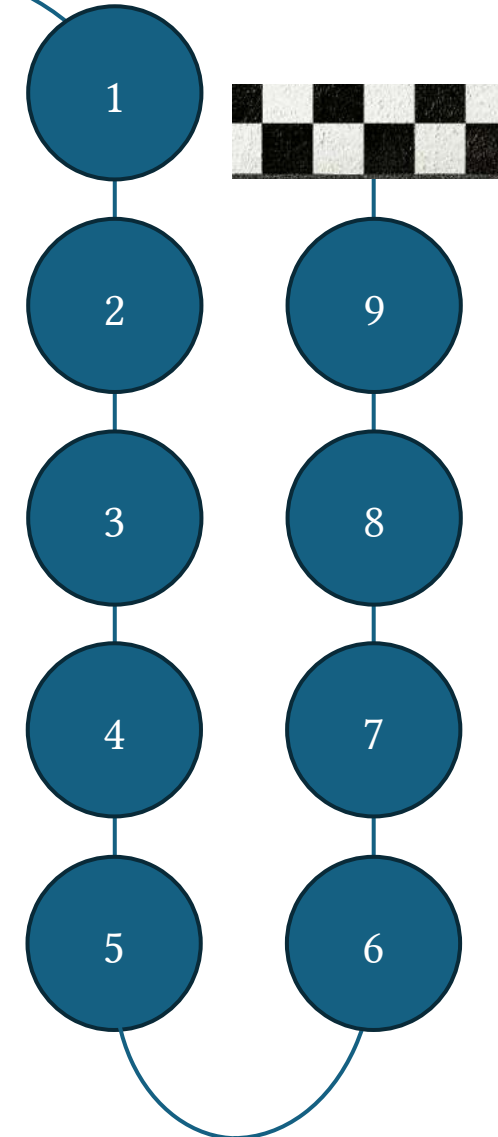
Caleb Pollreis



Overview



1. Intro / Dynamic Events
2. Driverless Roadmap
3. Why start early?
4. High Level System Architecture
 - Mechanical Integration
 - Electrical Integration
 - High Level Software Implementation
5. DSO – Driverless Systems Officer
6. Driverless Operations
7. DSF Submission
8. FSAE / FSG Resources
9. Why do this at all?



FSG Driverless 2023 Fastest Lap

<https://www.youtube.com/watch?v=9MWKDJJeAEDU>



Dynamic Events

Dynamic Event Markings

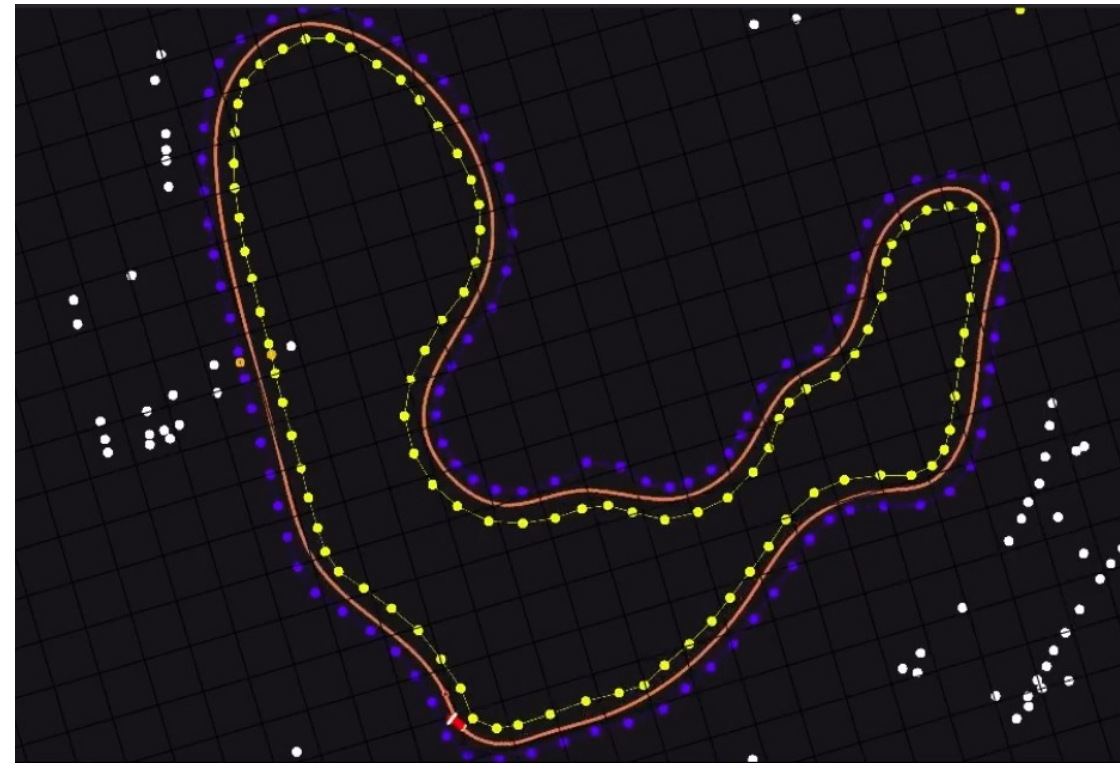
All Dynamic Events are marked by:

- Track Border, Left Side - Small Blue Cones
- Track Border, Right Side - Small Yellow Cones
- Entry and Exit Lanes - Small Orange Cones
- Before and after start, finish and timekeeping lines - Large Orange Cones
- Timekeeping equipment may be surrounded by additional cones outside of the track boundary

DD.1.2.4 No map data is given

DD.1.3.1 Details of Cones used to mark the Driverless Dynamic tracks:

	Cone Type	Stripe	WEMAS Part Number	
a.	Small Blue	Single White	400.000043.00.00	
b.	Small Yellow	Single Black	400.000013.01.10	
c.	Small Orange	Single White	400.000013.00.00	
d.	Large Orange	Dual White	307.610500.00.00	



Driverless Roadmap

SUMMARY TIMELINE

Year	Driverless Participation	Key Elements
2026	Optional	• Technical Inspection • Design • Up to 3 Dynamic Events
2027 - 2029	Optional	• Technical Inspection • Design & Cost • Up to 4 Dynamic Events
2030	Official	• Driverless Acceleration & Skid Pad events adopted into overall competition points ◦ Teams may opt-out ◦ Autonomous runs required for full scoring of 1000 points • Driverless Cup

- Vehicle must be able to operate in manual mode and driverless mode

Why Start Early?

- Driverless:
 - Drastic increase in system design complexity
 - “Working car > Optimized Car”
for all systems
 - Anything you don’t understand/Isn’t documented on the car currently is “complex”
- “Better is the enemy of good”

2) Learn on the job

Nobody tells you how to build a driverless racecar. There’s no book with step by step instructions, lecture slides, or professors that have 20+ years of experience that are advising you. Formula SAE is a crash course on being pushed off the edge of a cliff and learning how to fly.

3) You’re just as capable now as you will be 2 years from now

Transitioning from student to engineer requires starting to make your own design decisions and advocating for your own system. The best engineers I’ve met have a common trait: they’re not afraid to ask questions or look like they don’t understand something. *That’s how they learn.*

Being a software engineer in a hardware-heavy project is a unique experience. You’re not just writing code. You’re responsible for understanding mechanical and electrical systems you didn’t design, figuring out how different components interface and communicate, and often connecting all the pieces together. You spend time reading datasheets to understand sensor limitations, learning how signals propagate, and solving problems that span both hardware and software.

For example, from February to March, the EE⁰³ and software teams push to test our electrical system off the car by placing all our boards on a table and wiring them together for a full integration test. We call this process “EE on Bench,” where the final goal is to spin the motors that power the car with our custom battery. During that time, the software is actually mostly written, so I spend most of my time helping with assembly. I help make the wiring harnesses, assemble the boxes, and do full integration tests with the help from the electrical team.

High Level System Architecture



Software Integration
(High Level)

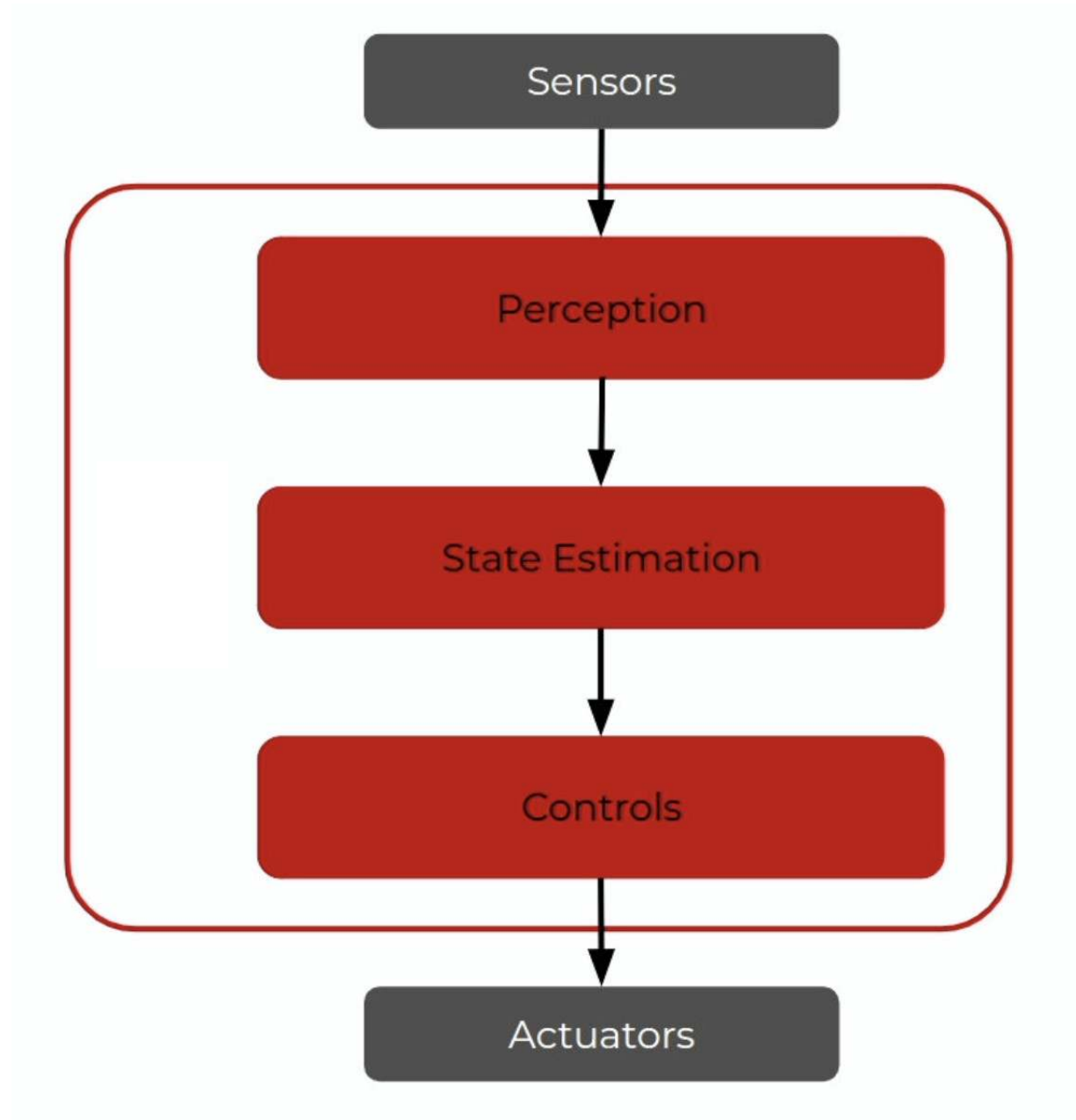


Mechanical Integration



Electrical Integration

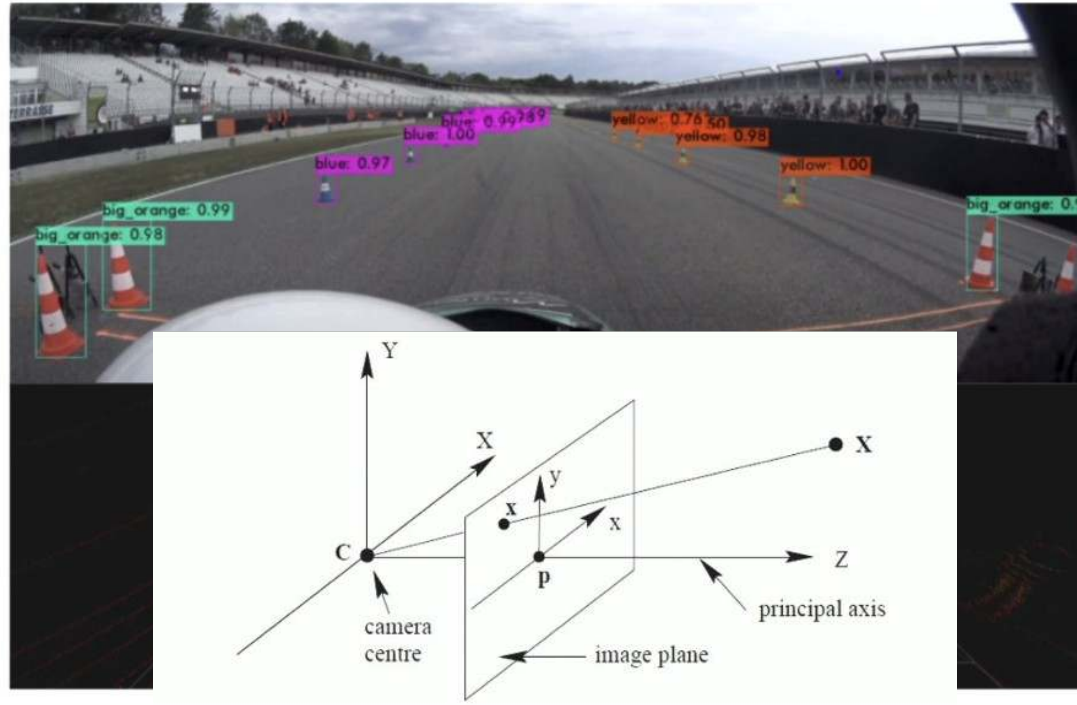
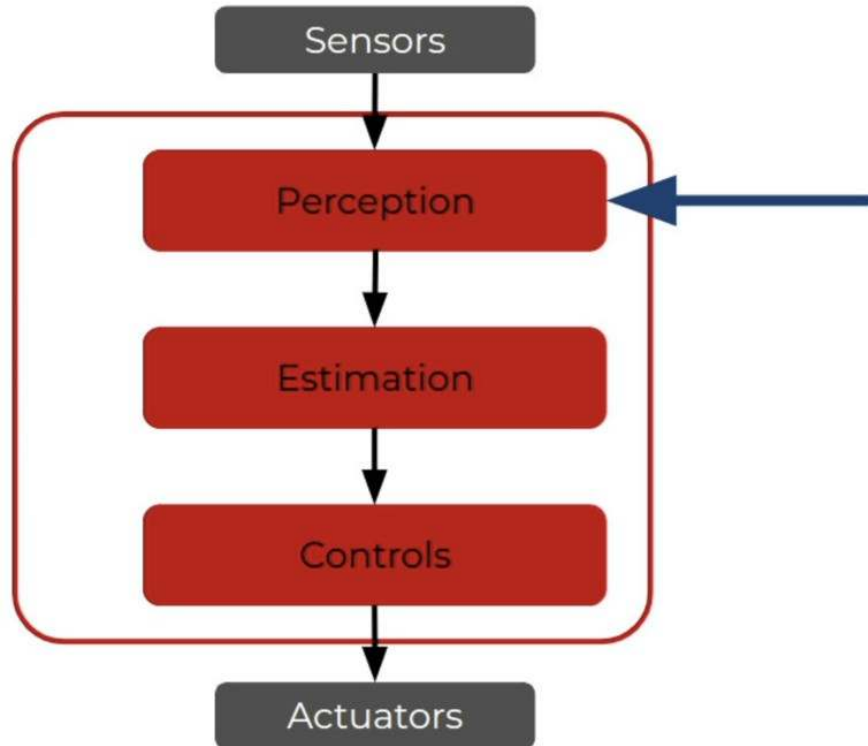
High Level Software Implementation



High Level Software Implementation

PERCEPTION

DETECTION AND LOCALIZATION OF ENVIRONMENTAL OBJECTS

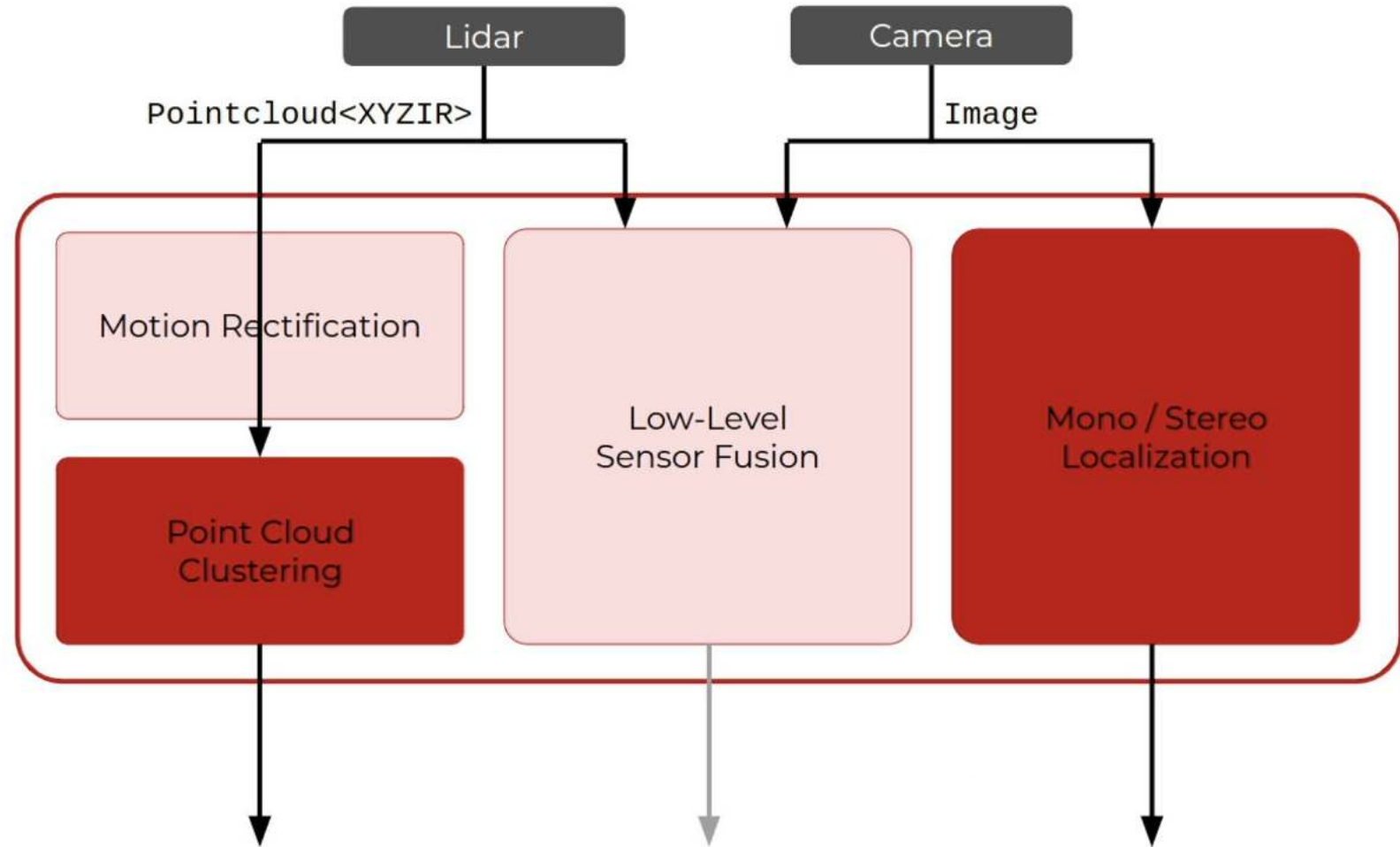
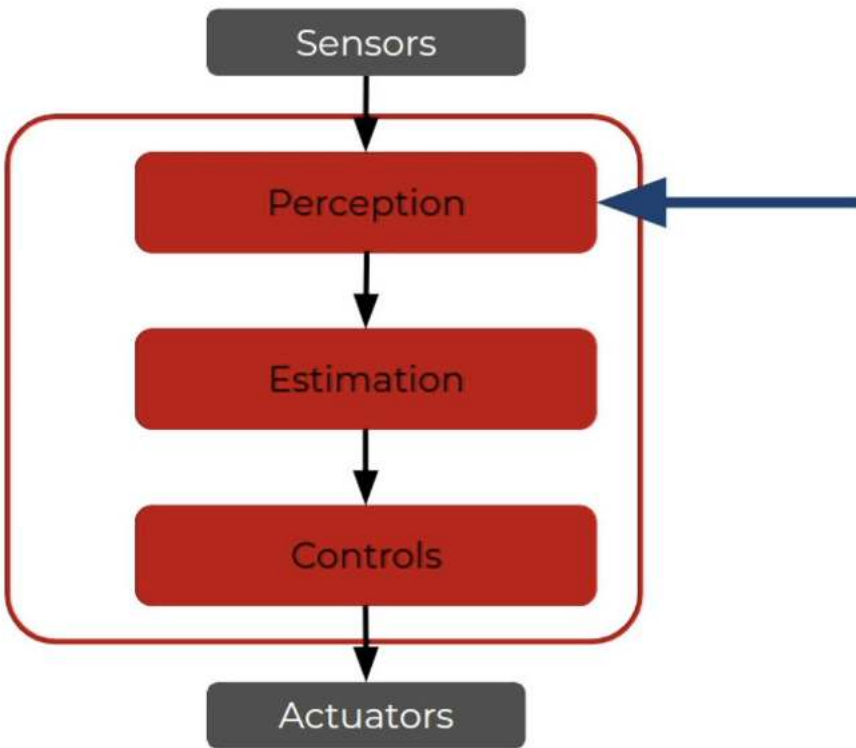


- Mix between LiDAR + Stereo Camera
- Sensor fusion (next slide) optimally but likely too complex starting out

High Level Software Implementation

PERCEPTION

DETECTION AND LOCALIZATION OF ENVIRONMENTAL OBJECTS

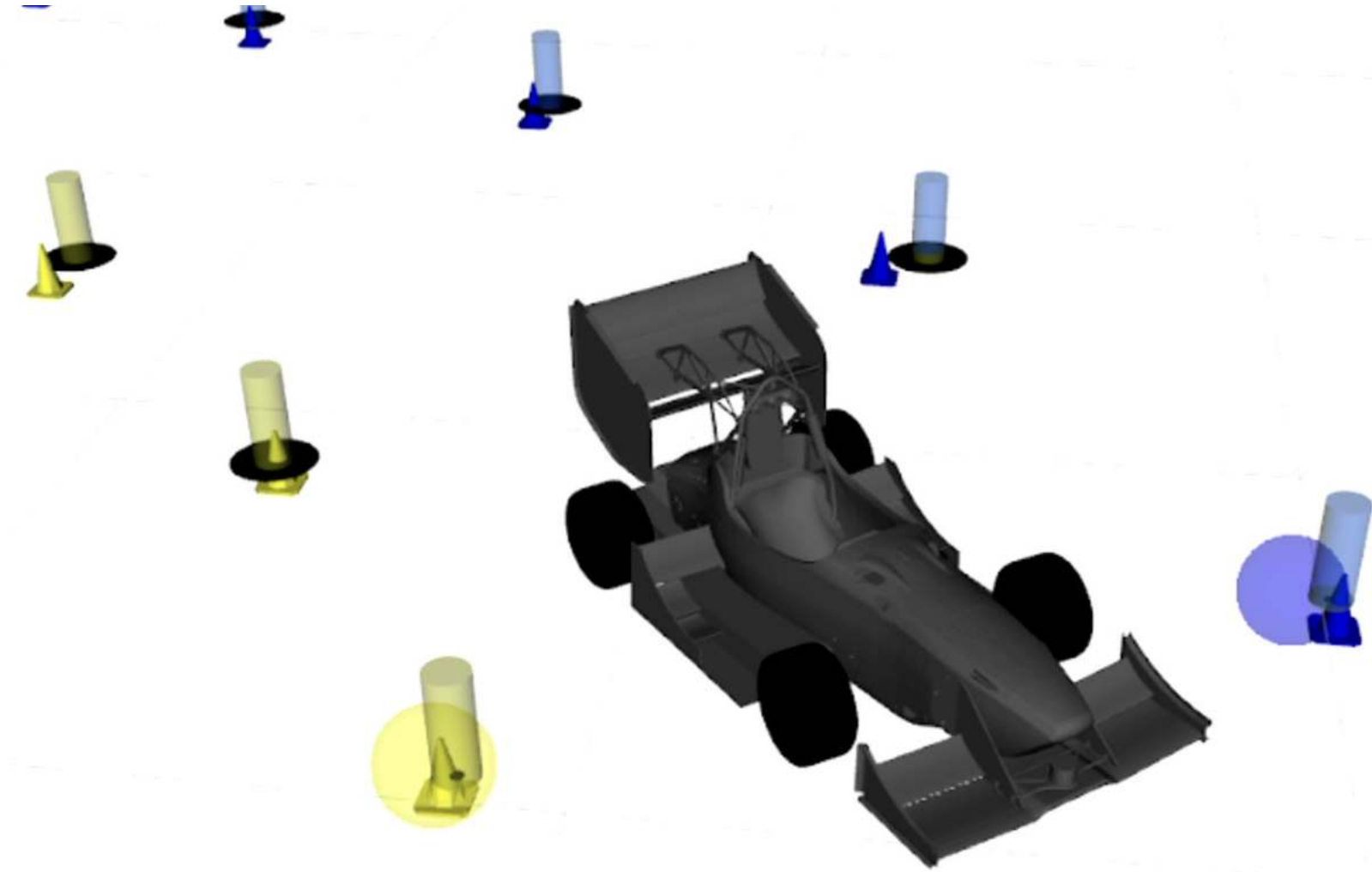
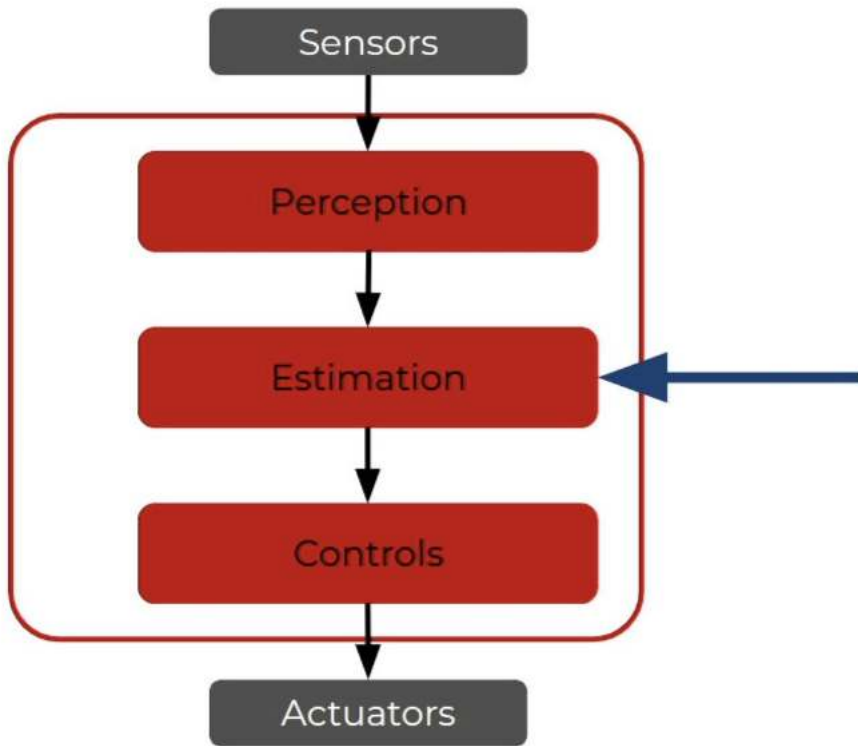


- Cone detection
 - Timestamps
 - Ref. Coordinate Frame
 - List of cones
 - Probability, Position etc.

High Level Software Implementation

ESTIMATION

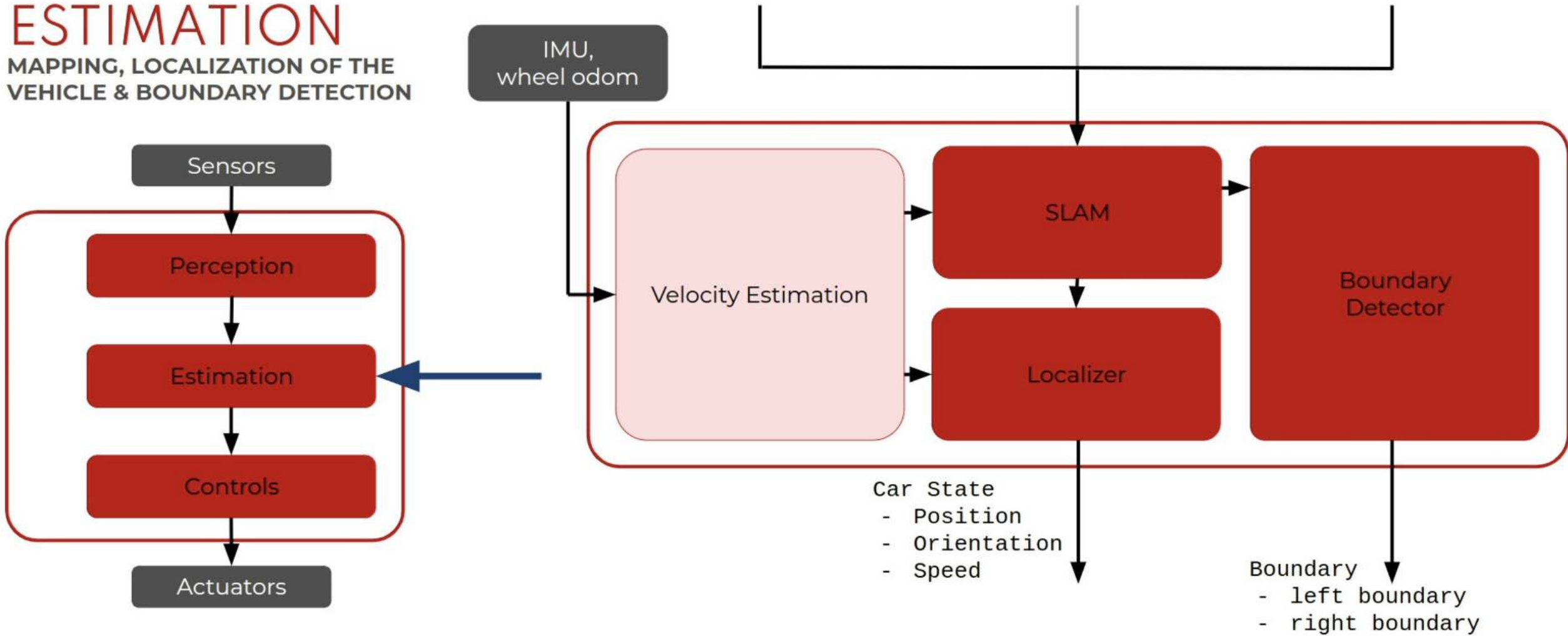
MAPPING, LOCALIZATION OF THE
VEHICLE & BOUNDARY DETECTION



High Level Software Implementation

ESTIMATION

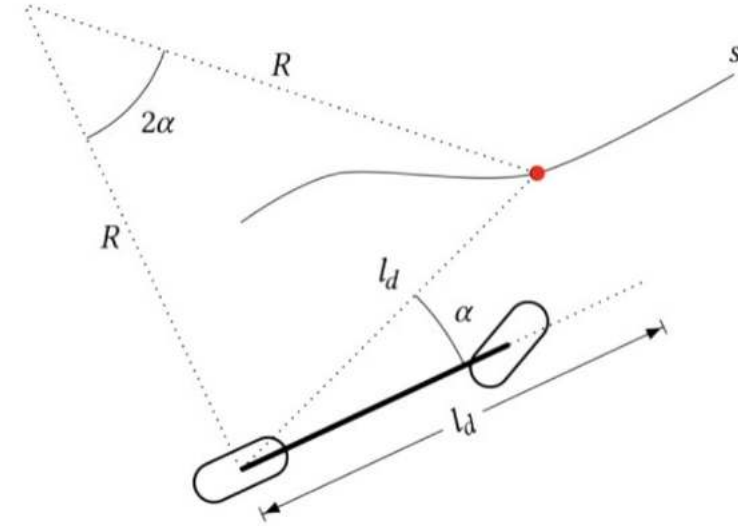
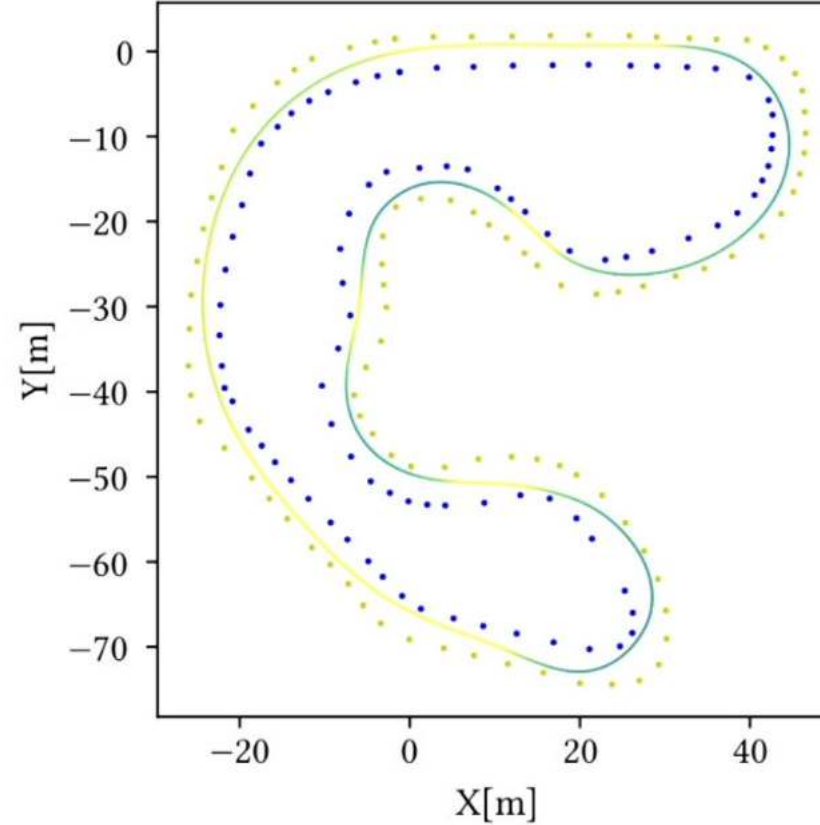
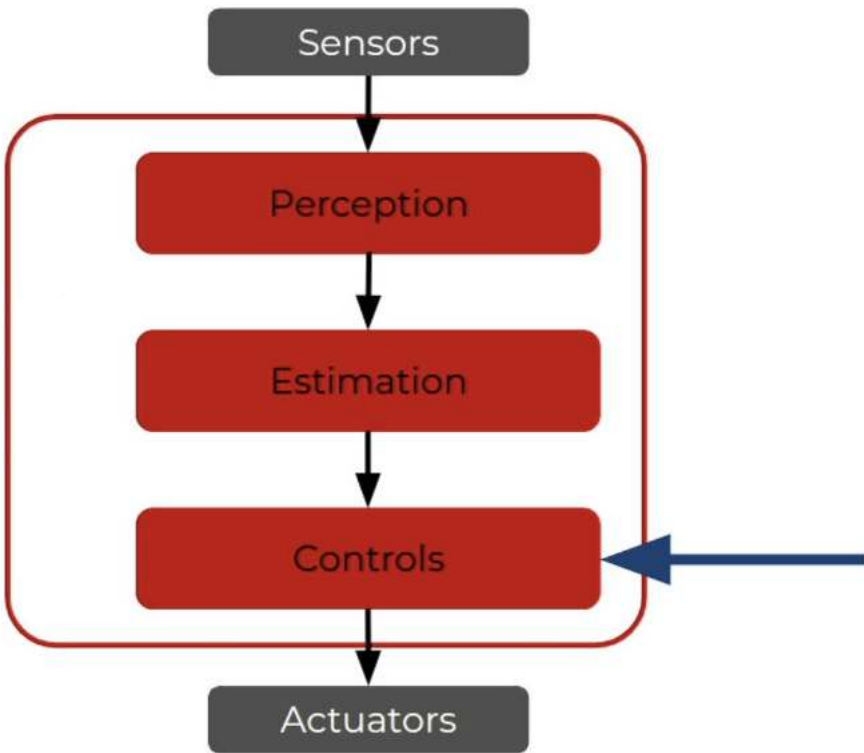
MAPPING, LOCALIZATION OF THE VEHICLE & BOUNDARY DETECTION



High Level Software Implementation

CONTROLS

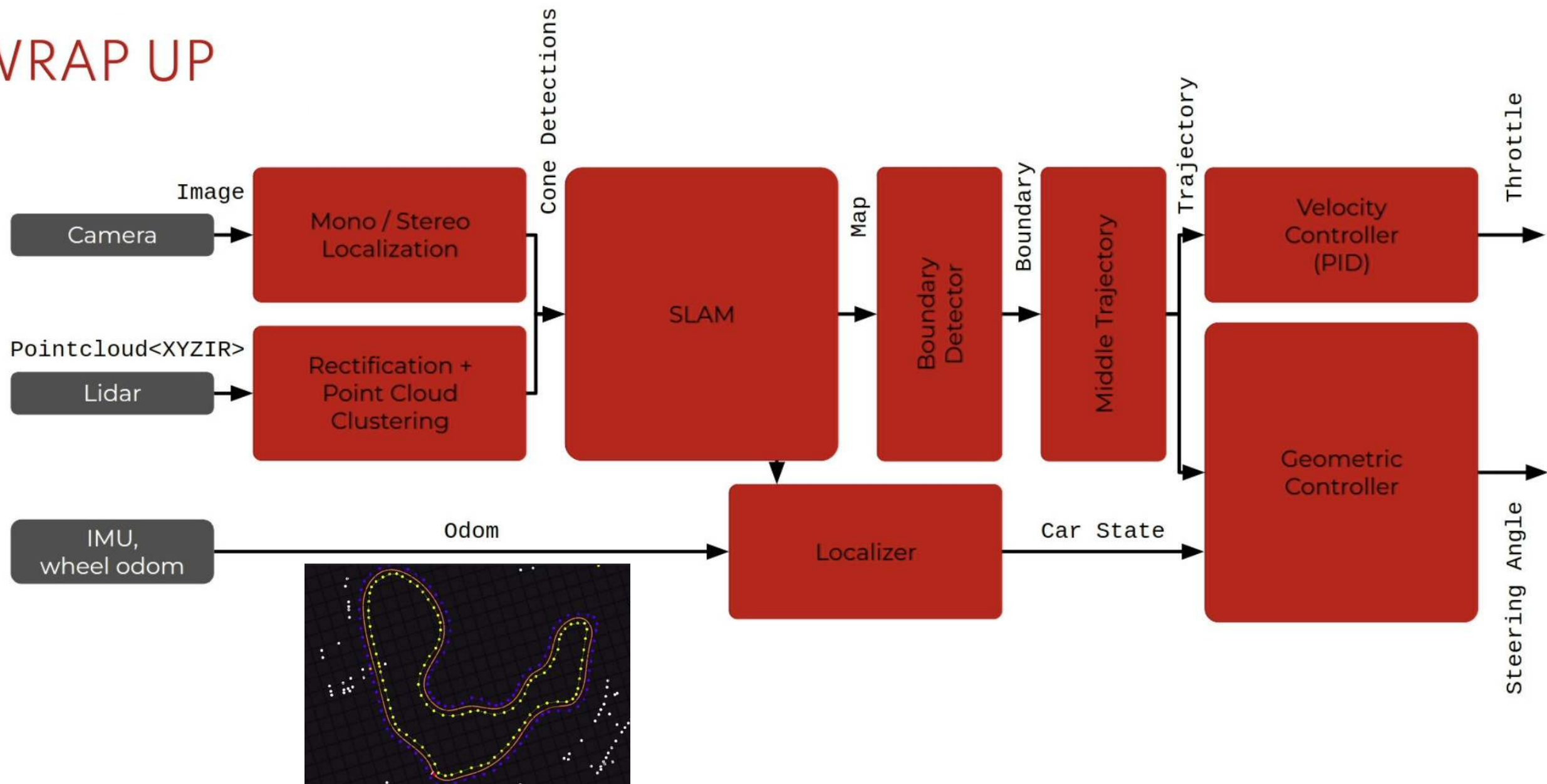
TRAJECTORY & CONTROLLING



- MPC Controller would be the best long-term solution
- Starting by finding the midpoint of the track is easier

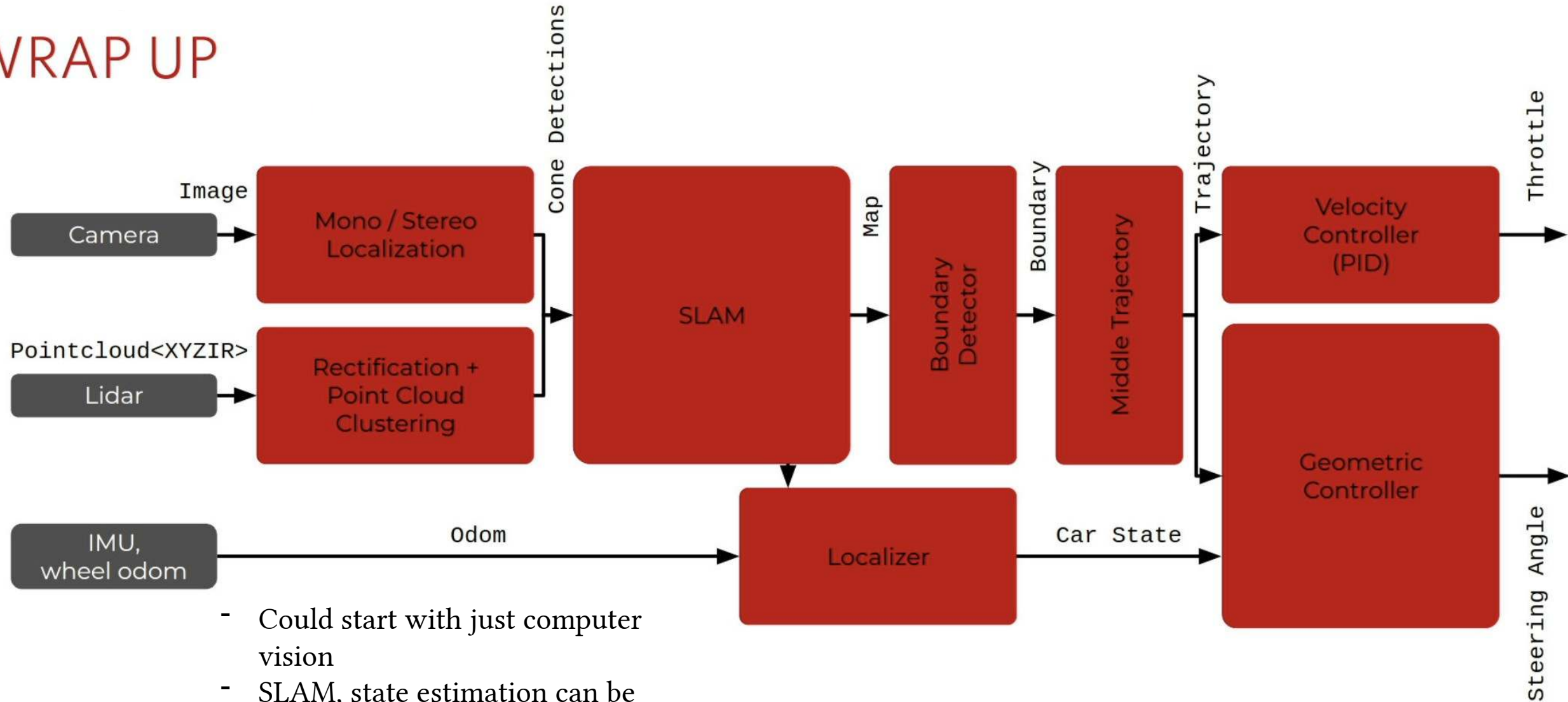
High Level Software Implementation

WRAP UP



Simplified Software Implementation

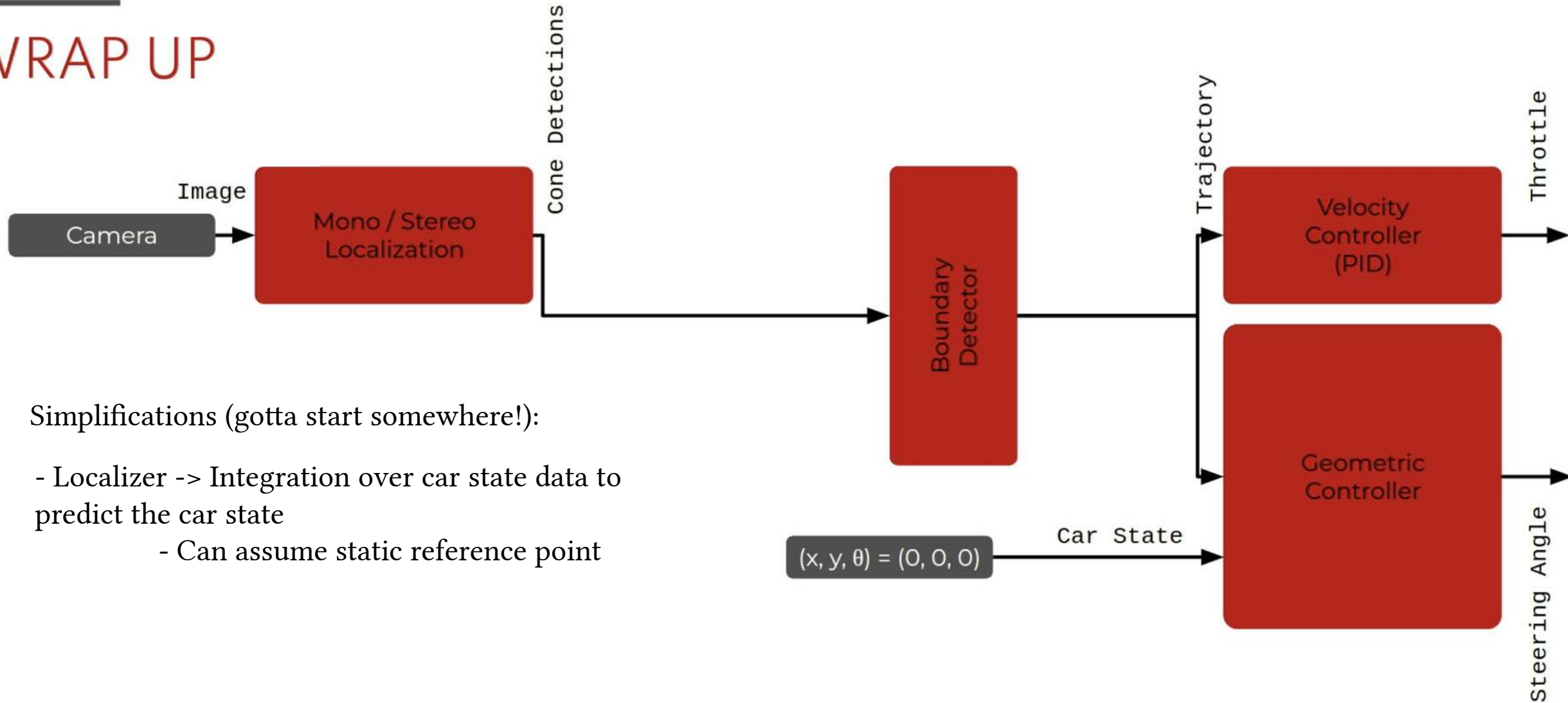
WRAP UP



- Could start with just computer vision
- SLAM, state estimation can be ignored to start with but required for competition

Simplified Software Implementation

WRAP UP



Simplifications (gotta start somewhere!):

- Localizer -> Integration over car state data to predict the car state
- Can assume static reference point

Breaking Down Simplified Approach



Boundary generator



(Geometric) steering controller

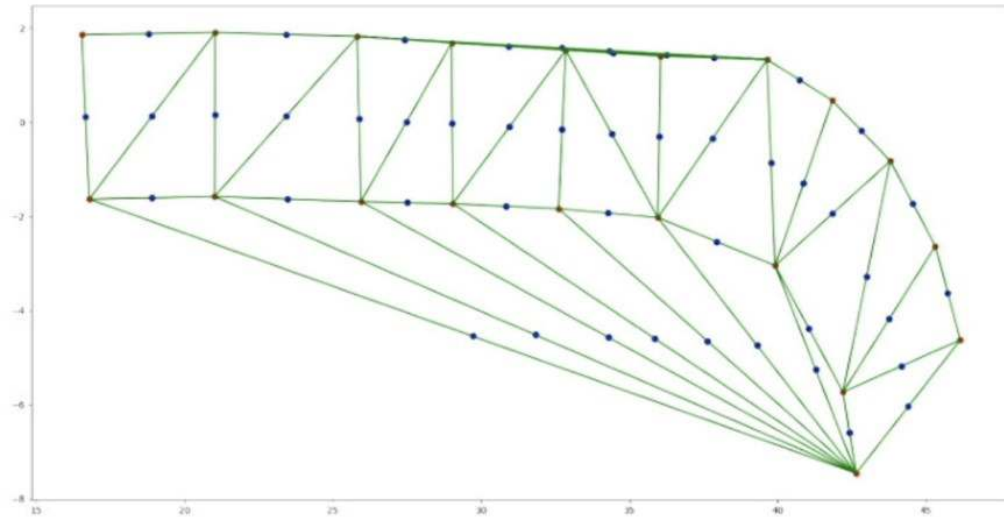


Velocity controller

Breaking Down Simplified Approach

BOUNDARY GENERATOR

LOCAL DELAUNAY TRIANGULATION

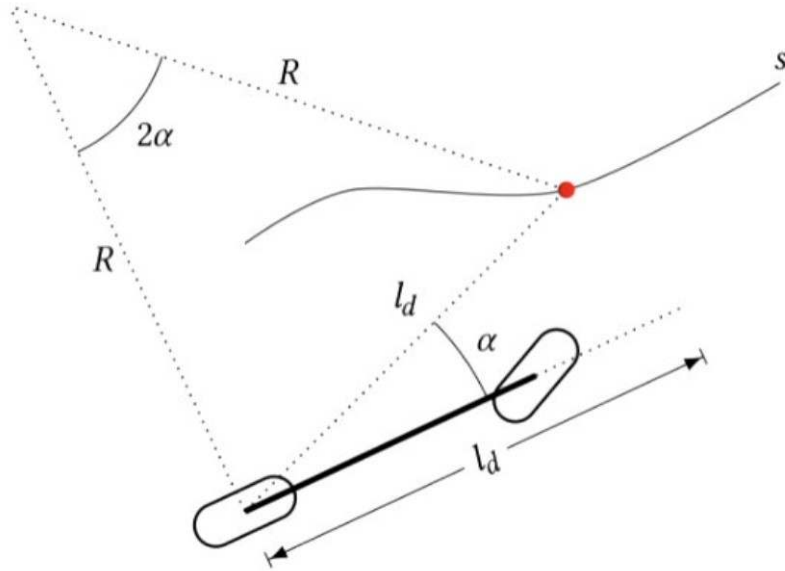


- Delaunay results in graph of triangles
 - Containing all **cones as nodes**
 - All neighbouring **cones** connected through **edges**
 - No **edges** are crossing, only triangles exist
- Connect **middle points** of all **edges** of a **cone**, each
 - Each **middle point** is either part of
 - the track
 - boundary
 - none
 - resulting in another graph
- The easiest criteria for a **point on the track**: **Edges** belong to **cones** of different class (e.g. blue, yellow)
- Working only on sight, in the frame of the vehicle

Breaking Down Simplified Approach

GEOMETRIC CONTROLLER

ONE POSSIBILITY: PURE PURSUIT

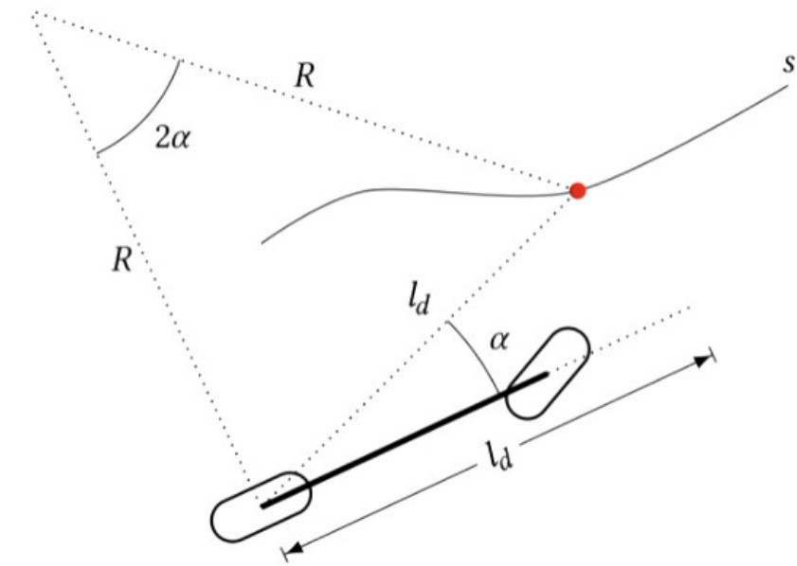


- Input: Middle trajectory (list of points) as reference
- Params
 - Length of vehicle
 - Lookahead distance / radius
- Preparing input data (optional)
 - Densify trajectory for better sampling
 - (Interpolate the trajectory)
- Generate output
 - Select **target point** based on lookahead distance
 - Steering angle results in angle of target point to the vehicle's longitudinal axis

Breaking Down Simplified Approach

VELOCITY CONTROLLER

PID



- Hold a static velocity
 - Issue acceleration
 - Issue deceleration
 - Target velocity, keeping the rules in mind (no DV Cup):
 - D 5.5.1: Runs with a time without penalties > 25 s will be DQ.
 - D 5.5.2: Worst scoring:
$$75 / \text{NUM_OF_ALL_ACCEL_FINISHERS}$$
- Keep in mind, a PID can be used without competing in the DV cup / DV events
- SLAM (or some other algorithmic approach) would be required to run, and perform well, on an unknown track

Enabling Work From Anywhere

AMZ sim from 7 years ago
(best sim at the time, outdated):

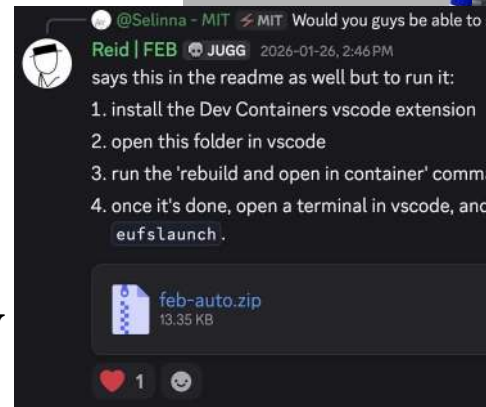
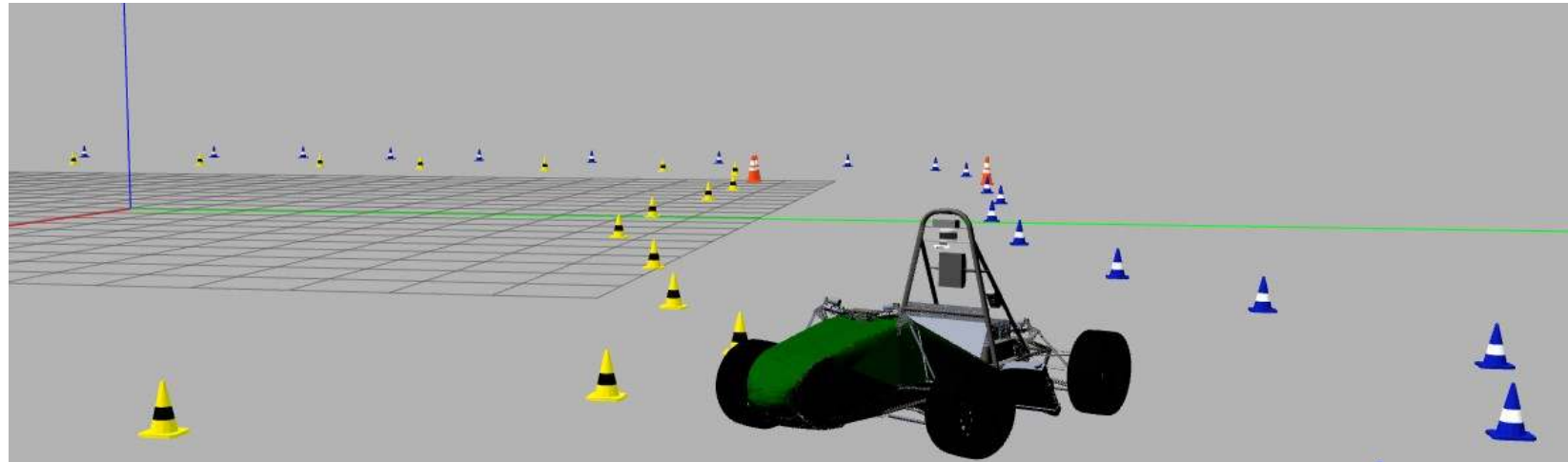
<https://github.com/AMZ-Racing/fssim>

EUFS path planning sim that's more up to date:

https://gitlab.com/eufs/eufs_sim

Formula Student official sim for LiDAR and camera realistic tests:

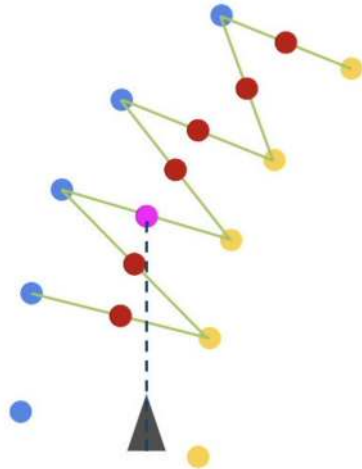
[click here](#)



- The team has been limited without a sim or any method of remote work
- There's a working version kindly provided by the DV discord -->

Good Software Design Means ...

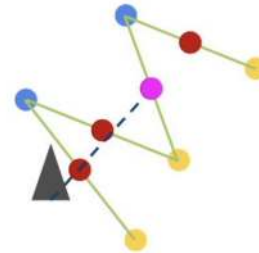
- Understanding Design Requirements and Constraints
- “What components does the software interface with?”
- “Who is going to be using the code?”
- “What is their skill set?”
- Latency?
- Memory?
- Power Draw?



Rough Hand Calc

Straight-line 40kph to pass EBS test
(ignoring accel.)

- If you sample inputs at (example) $f = 5\text{Hz}$
- $T = 200\text{ms}$ latency (+ any added latency from other systems)
- $d = vt = (11.11 \text{ m/s})(0.2 \text{ s}) = 2.22 \text{ m}$
 - Discrepancy from sampling to actuation(s)
- Too slow to react to oncoming turns @ high speeds as well



Driverless System Outline

- Driverless System
- Needs more computing power than current STMs, needs internal storage for ROS bags
 - NVIDIA Jetson Orin (good balance of cost to compute power for this application)
 - Need something more robust than a DEV kit on the car, could put internals in an enclosure
- OS: Ubuntu 20.02
- Framework: ROS2 ($\geq$ Galactic)
- Cost:
 - Really depends on what we can get sponsored
 - PTx Trimble -> informally offered to give us camera(s) + LiDAR sponsored
 - Jetson -> Not sure yet, likely sponsored as people move to Jetson Thor



+



+

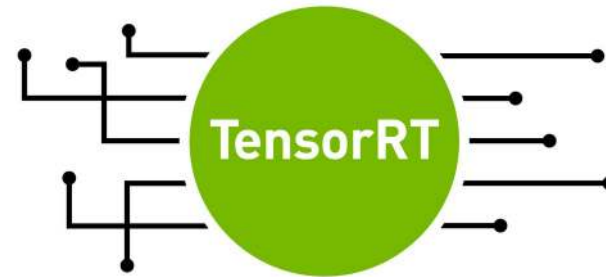


Driverless System Outline

ONA



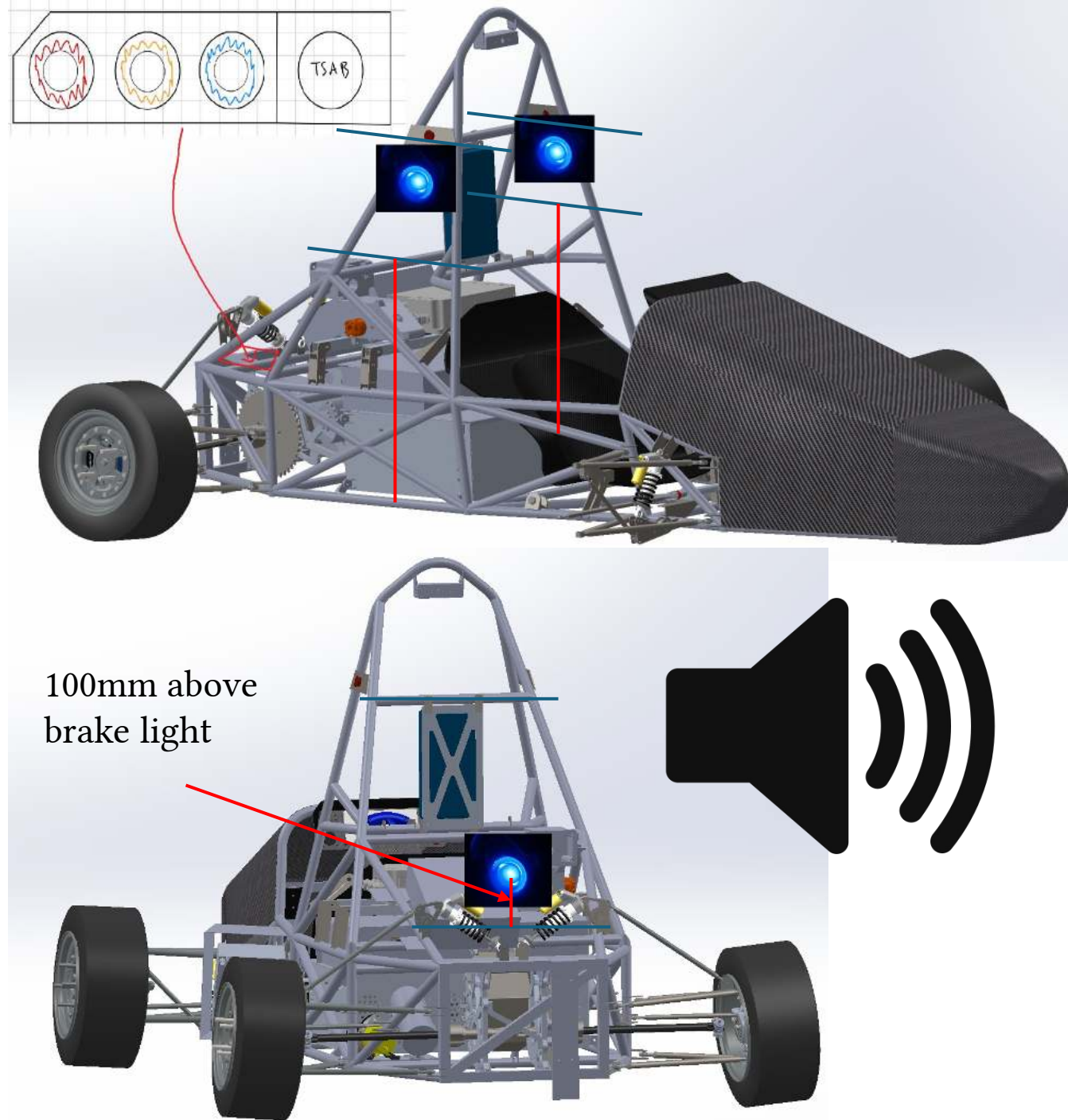
ROS2



And many more!

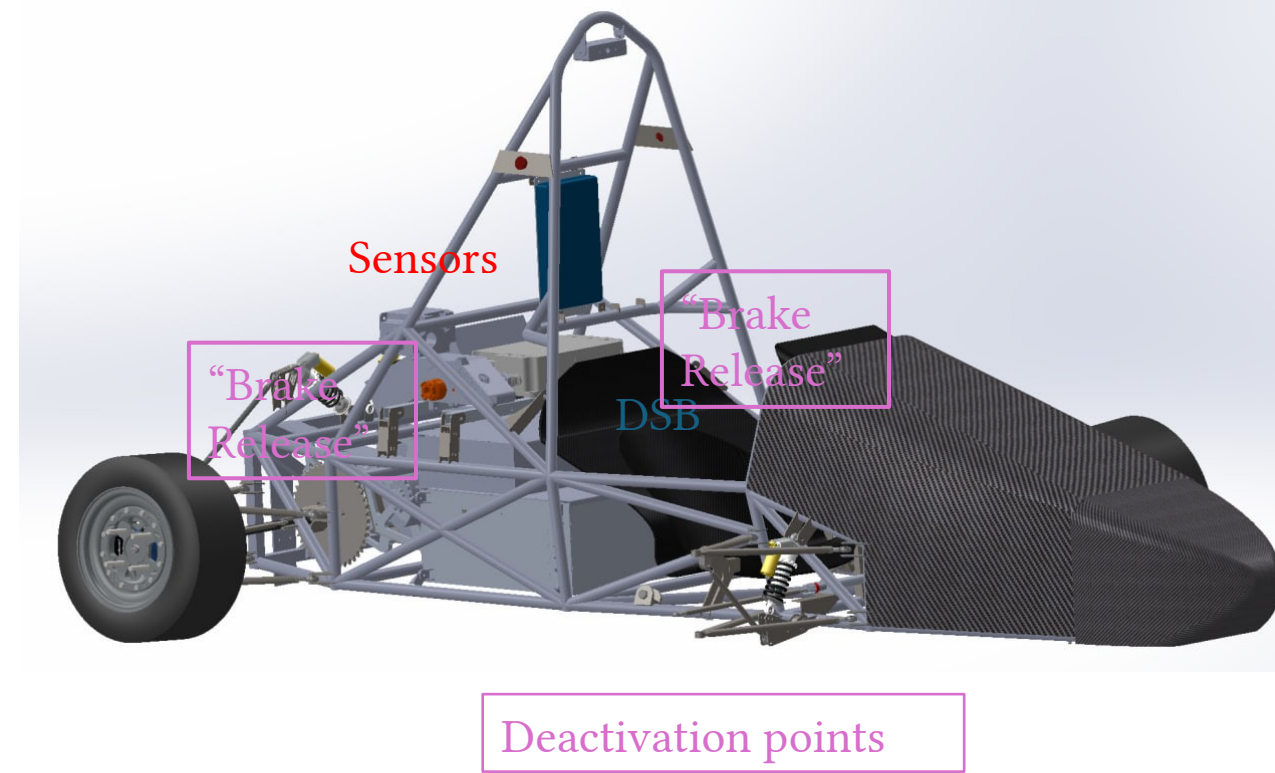
Electrical – Controls + Indicators

- DSMS – Driverless Systems Master Switch
 - Driverless Key
- Tractive System Activation Button
 - Puts vehicle TSA from outside vehicle
- DSSI - Driverless System Status Indicators
 - 3 lights -> 2 on main roll hoop, 1 above brake light
 - Shows Driverless Systems Status when GLVMS and DSMS are ON
- Driverless Alert Sound -> For Emergency State
- Driverless Mission Indicator (separate slide)



Mech – Sensor Mount + Steering + DSB

- Manual Steering still possible
- Actuator Decoupling
- Sensor Mounting
- Brake System(s)
 - DSB -> Driverless System Brake
 - May be a part of the hydraulic braking system
 - MAX 2 deactivation points (1 near DSMS OR top side of vehicle)
 - Works without electrical power
 - Needs to build pressure before going driverless is ready
 - Manual Braking still possible



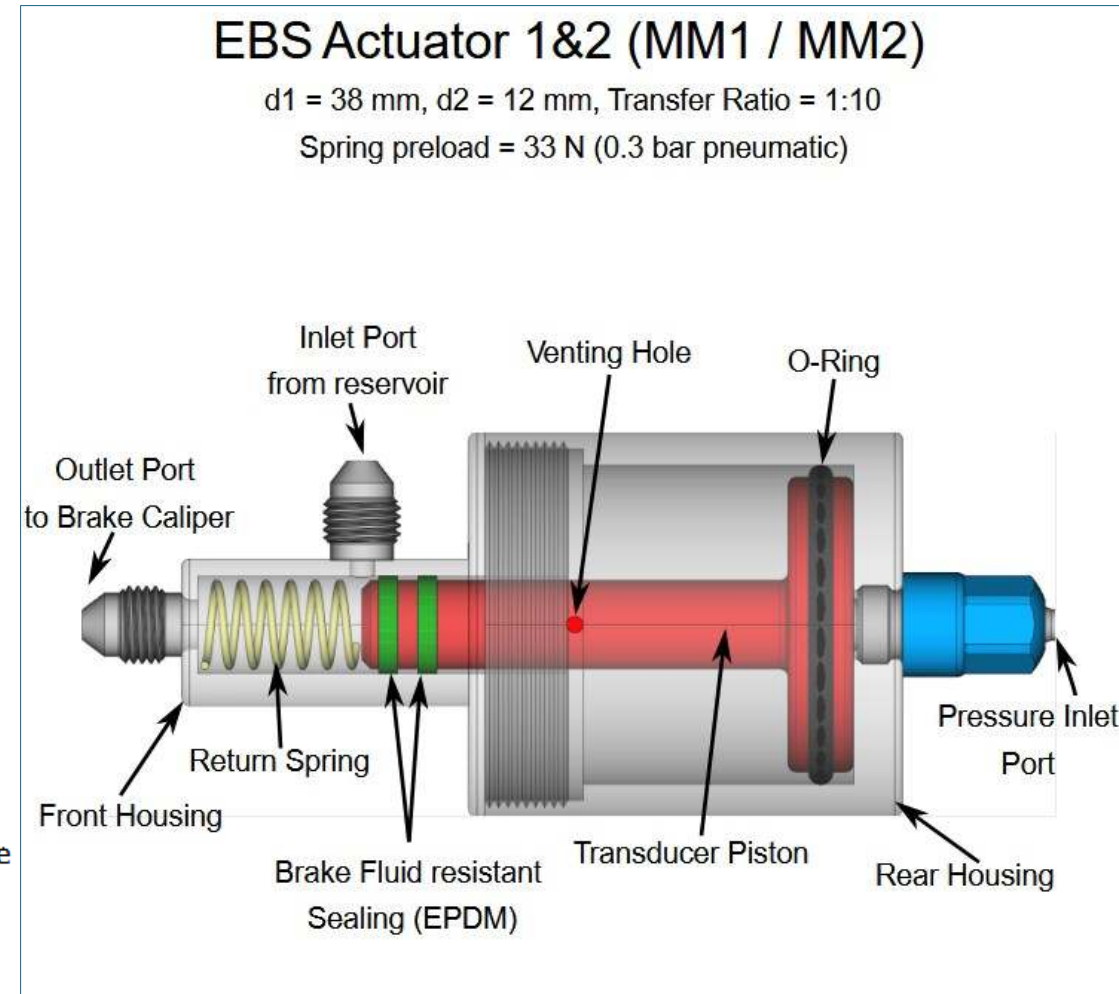
Mech – EBS (Emergency Brake System)

- EBS:
 - Only use passive systems (mech. energy storage)
 - GLVMS + DSMS + Remote Stop Relay (RSS) + Emergency Brake System Relay
 - All supplied directly, no delay
 - EBS Relay -> Start braking when shutdown circuit open
 - Car enables EBS when ... (see rules)
 - Failure of above signals (or power loss)

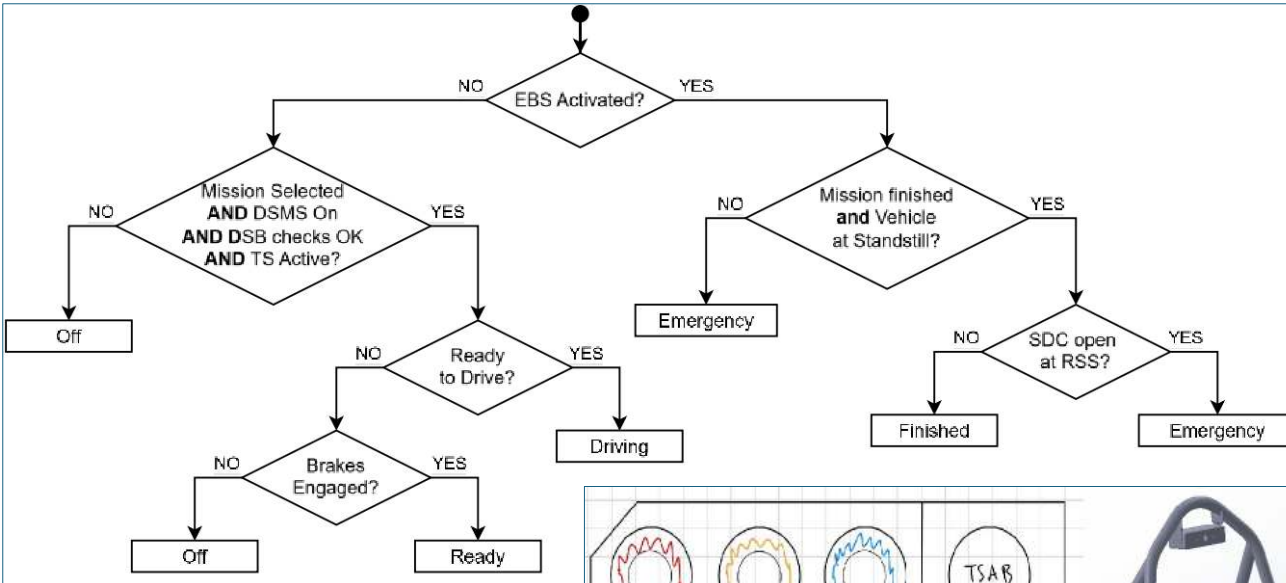
DT.3.2.5 Emergency Brake Maneuver

The Emergency Brake System must decelerate the vehicle and stop vehicle motion

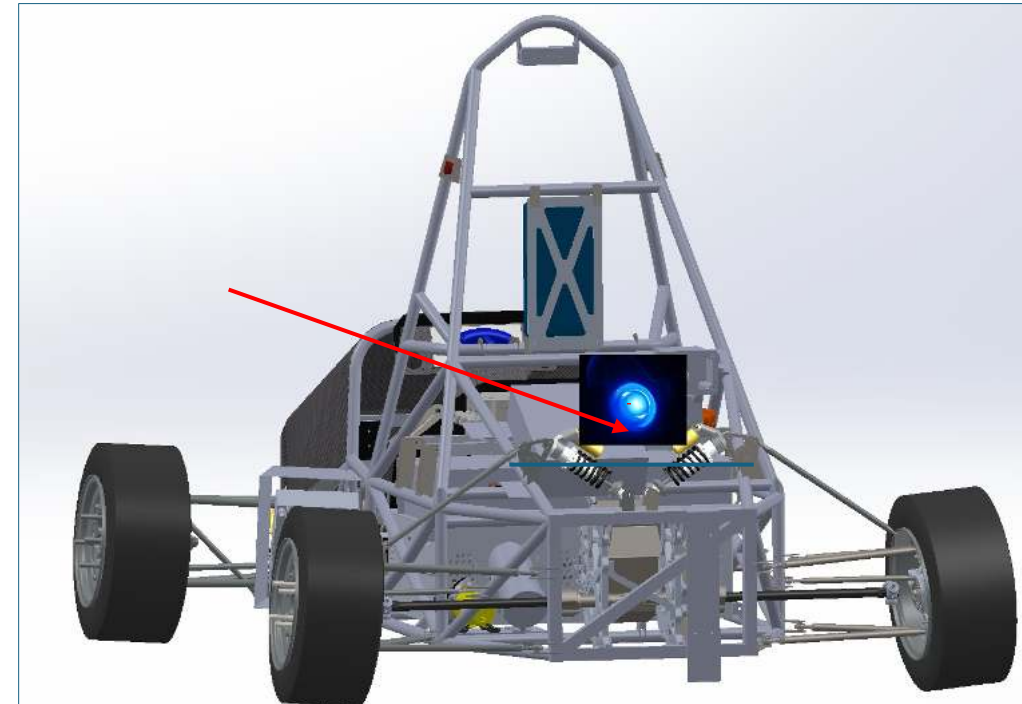
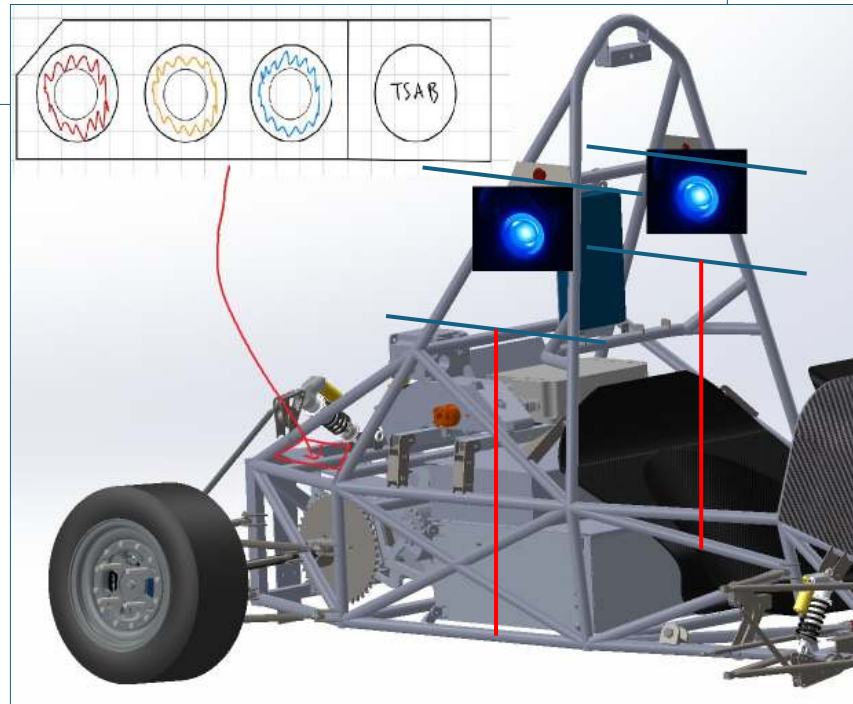
- The system reaction time, the time between opening of the Shutdown Circuit and the start of the deceleration, must be 200 ms or less
- The average deceleration must be more than 10 m/s^2 under dry track conditions
- In case of a single failure the EBS should achieve at least half of the performance
- While decelerating, the vehicle must stay in a stable driving condition



Driverless Systems Operations



	Driverless System Status	DSSI Indication
a.	Off	Off
b.	Ready	Yellow Continuous
c.	Driving	Yellow Flashing
d.	Finished	Blue Continuous
e.	Emergency	Blue Flashing



Driverless Systems Operations

DO.1.2 Driverless Missions

DO.1.2.1 The Driverless System must at least implement the following missions:

- Inspection **DO.1.3**
- Manual Driving
- Emergency Brake System Test
- Acceleration
- Skidpad
- Autocross

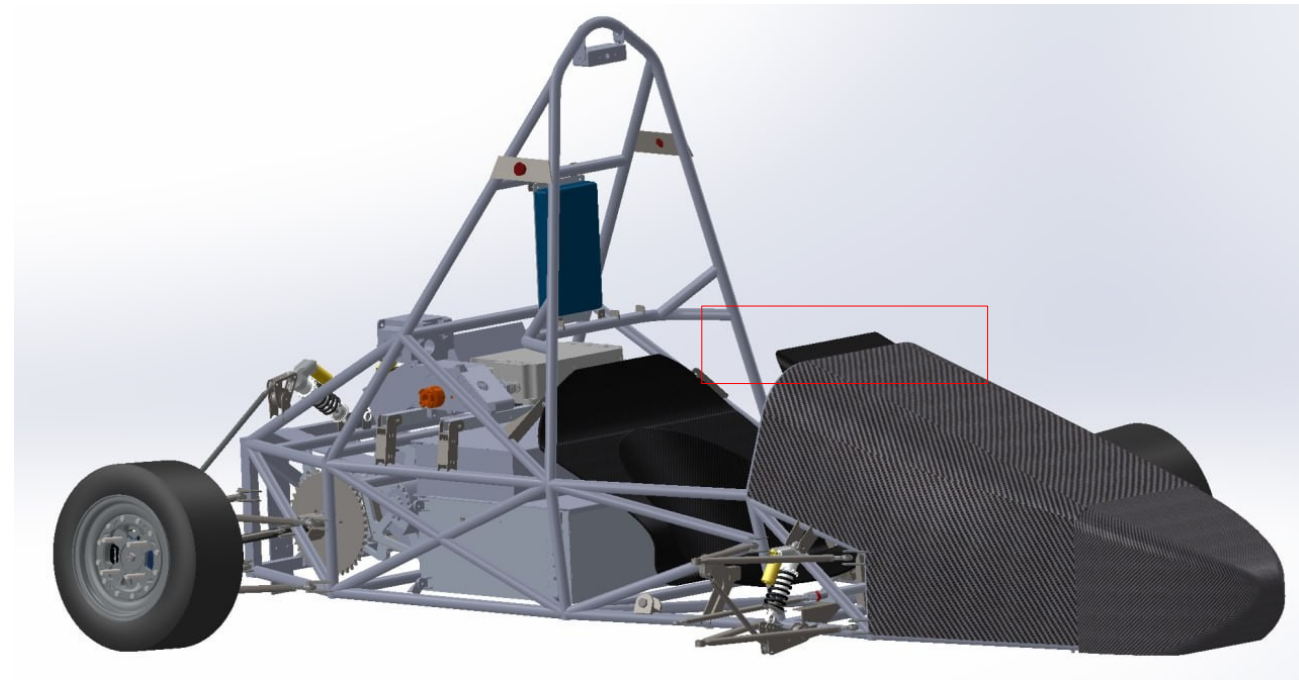
DO.1.3.2 The Inspection Mission must:

- a. Slowly spin the drivetrain
- b. Slowly operate the steering system with a sine wave pattern
- c. Transition to Driverless System Finished after 25 sec to 30 sec

DO.1.4 Driverless Mission Indicator

The vehicle must have a Driverless Mission Indicator that must:

- a. Show the selected Driverless Mission
- b. Be easily readable
- c. Be located as part of the dash or located near the DSMS



- Dash display could be updated to switch to a “driverless” screen on some trigger (haven’t looked into it yet)
- Or we switch it manually

Driverless System Inspections

DRIVERLESS SYSTEM INSPECTION

Inspection Required

To operate in Driverless Mode, the vehicle must pass the three:

- Technical Inspection **IN.3.1**
- Driverless Inspection **IN.3.2**
- Emergency Brake System Test

DI.1.2

Inspection Items

Bring these to Driverless Inspection:

- Data sheets for all sensors
- Documents which prove that all sensors meet local legislation
- Remote Stop Remote Control
- Tools needed for Driverless Inspection
- Copies of any submitted Rules Questions with the received answer

DI.2 EMERGENCY BRAKE SYSTEM TEST

DI.2.1 Objective

The Emergency Brake System is tested dynamically to make sure performance obeys **DT.3.2.5** requirements

DI.2.2 Conduct

DI.2.2.1 The Emergency Brake Test will be performed in a straight line marked with cones similar to Acceleration

DI.2.2.2 During the Emergency Brake Test:

- a. The vehicle must accelerate in Driverless Mode to 40 km/h minimum in 20 m or less
- b. The Remote Stop starts at a specific point
- c. The vehicle must come to a safe stop in 8.5 m or less

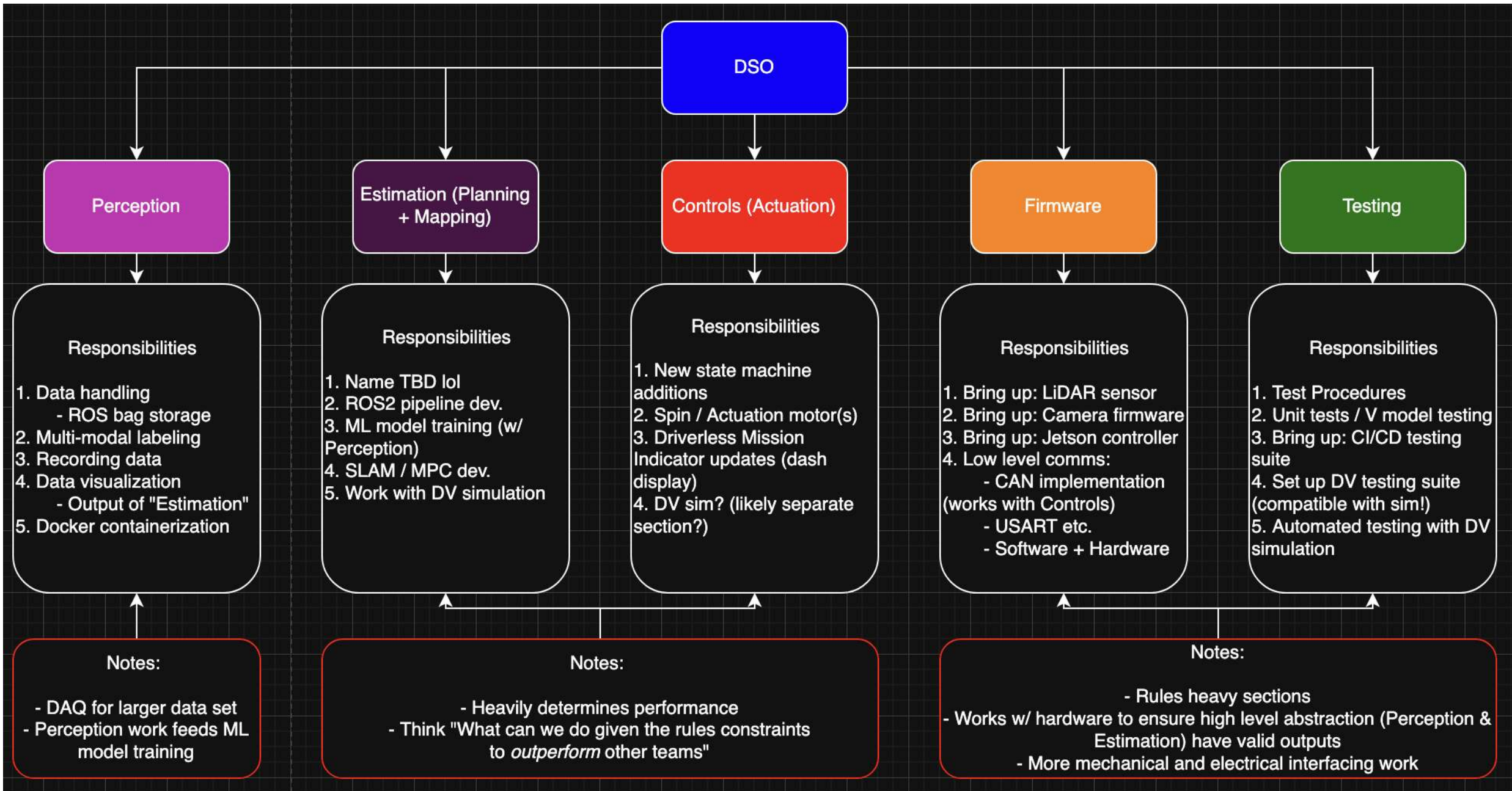
DI.2.2.3 If track conditions are wet, the stopping distance may be scaled by the officials

DI.2.2.4 The Emergency Brake Test is done after other parts of Technical Inspection are completed

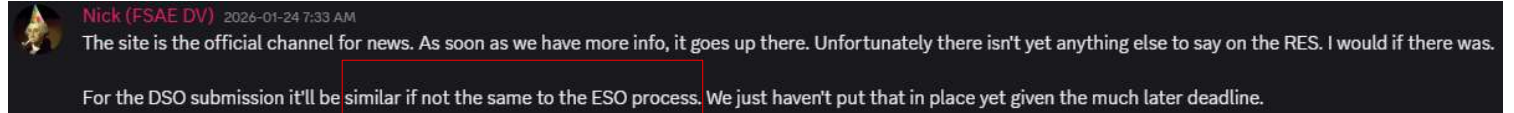
DSO – Driverless Systems Officer

- Responsible for Driverless System
 - Must not be a driver
 - Must be with car when driverless system is installed
 - DSF, Driverless Video, etc.
- Only person to:
 - Let the vehicle operate in manual mode
 - Declare Driverless System “safe”
 - Use the Driverless Key (on DSMS)

Software Team Structure



DSF – Driverless Systems Form



Submission Outline:

1. Actuator Power Supply Schematic
 2. Continuous Monitoring Flowchart
 3. Startup Checks Flowchart
 4. High Level Block Diagram
 5. Pressure System Overview
 6. Mechanical Assembly Drawings
 7. CAN DBC
- Each of the above has 2 sections:
 - System Requirements Checklist
 - Technical Content

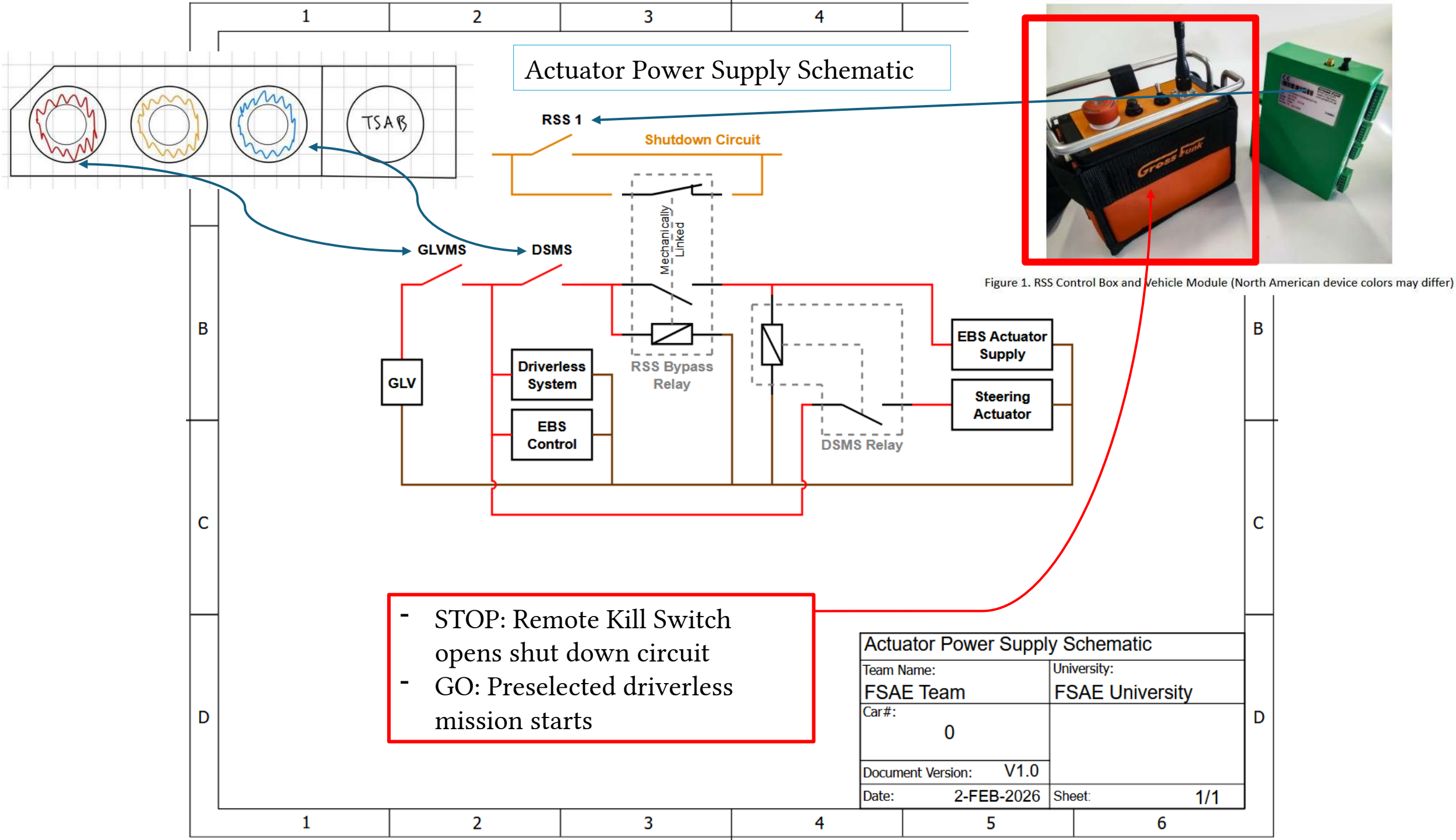


Electrical – Remote Stop System (RSS)

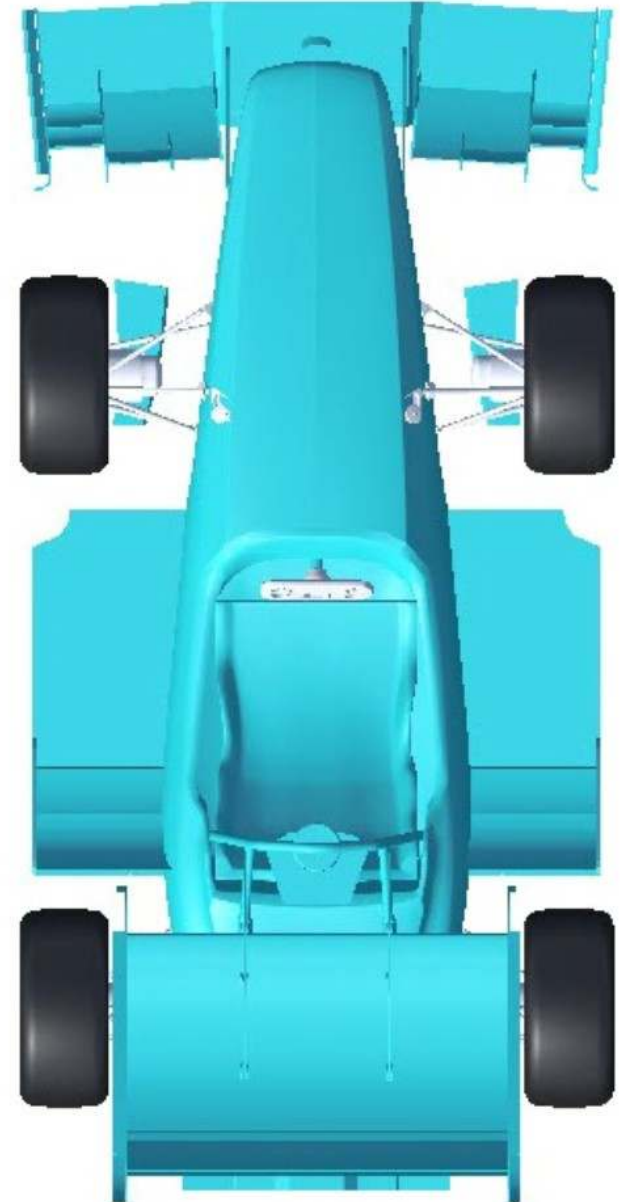
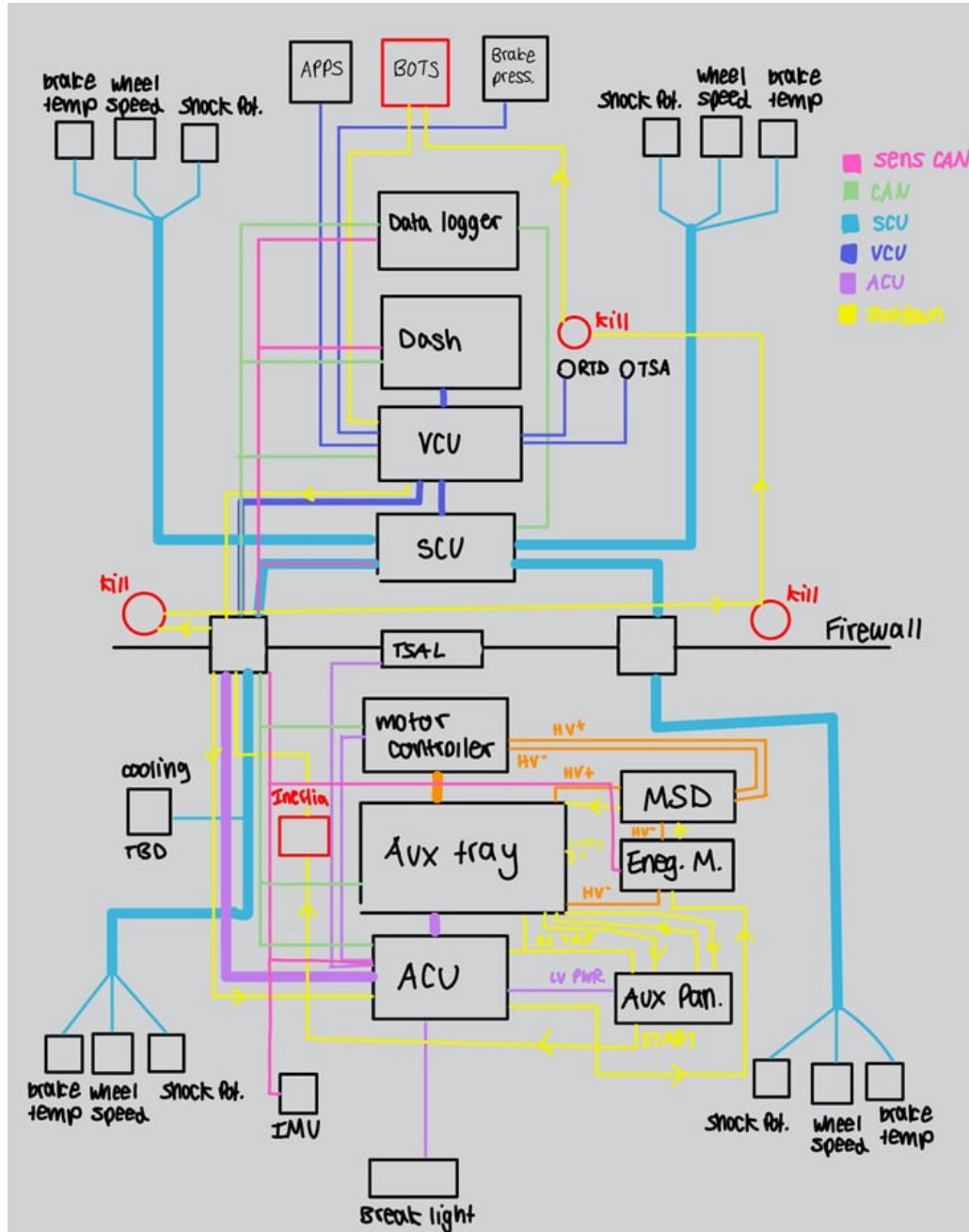
- RSS:
 - Pre-determined module by SAE
 1. Vehicle Module Box
 2. Control Box
 3. Remote Stop Relay
 - Bypassed when in “Manual Mode”
- Antenna of RSS mounted without being obstructed
- More on next 2 slides



Figure 1. RSS Control Box and Vehicle Module (North American device colors may differ)



ePBR26 Wire Harness



Actuator Power Supply w/ ePBR26 Overlay

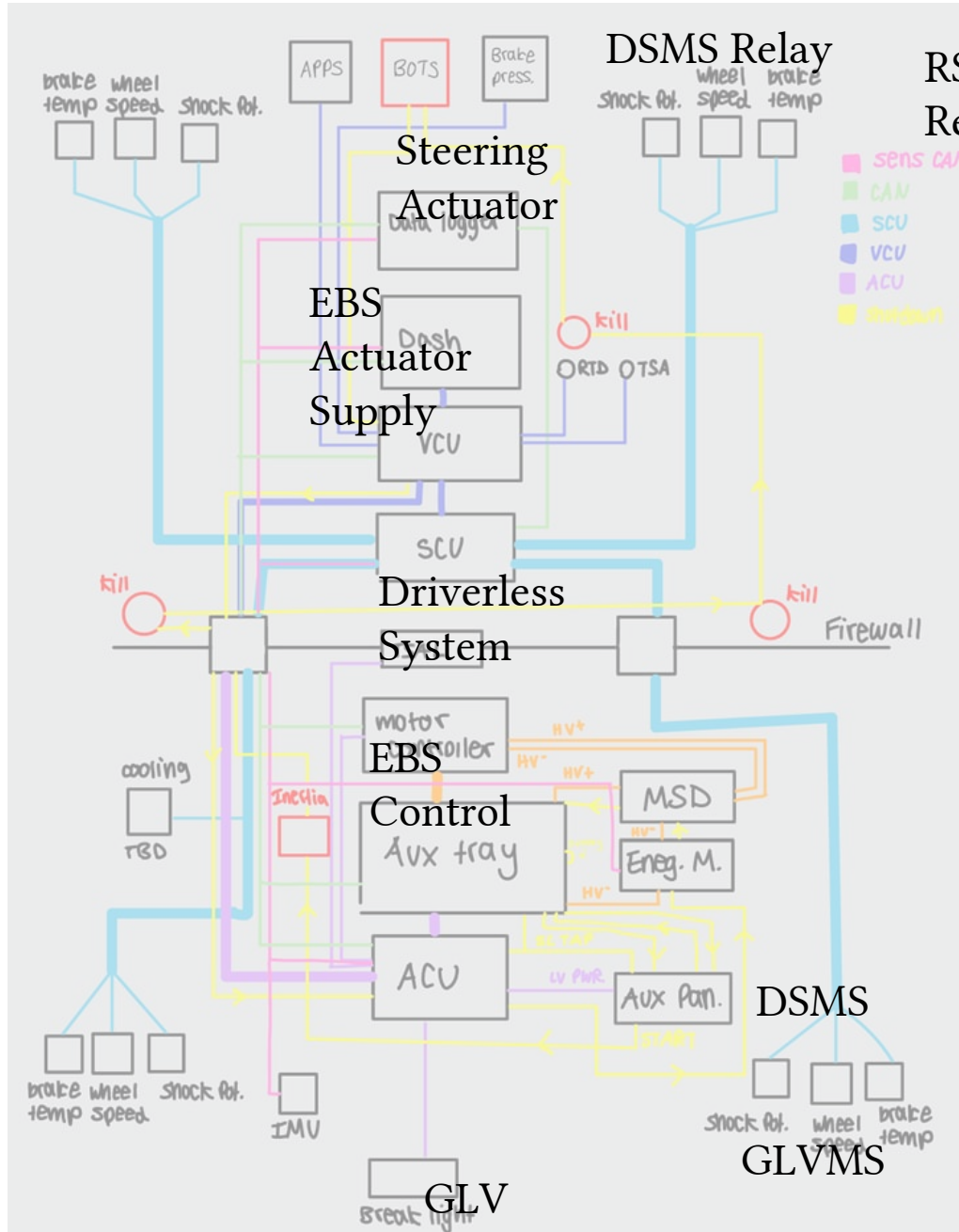
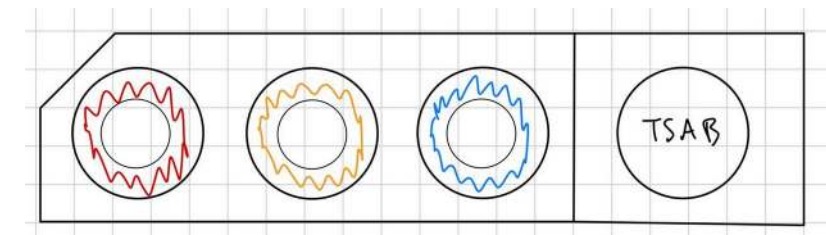
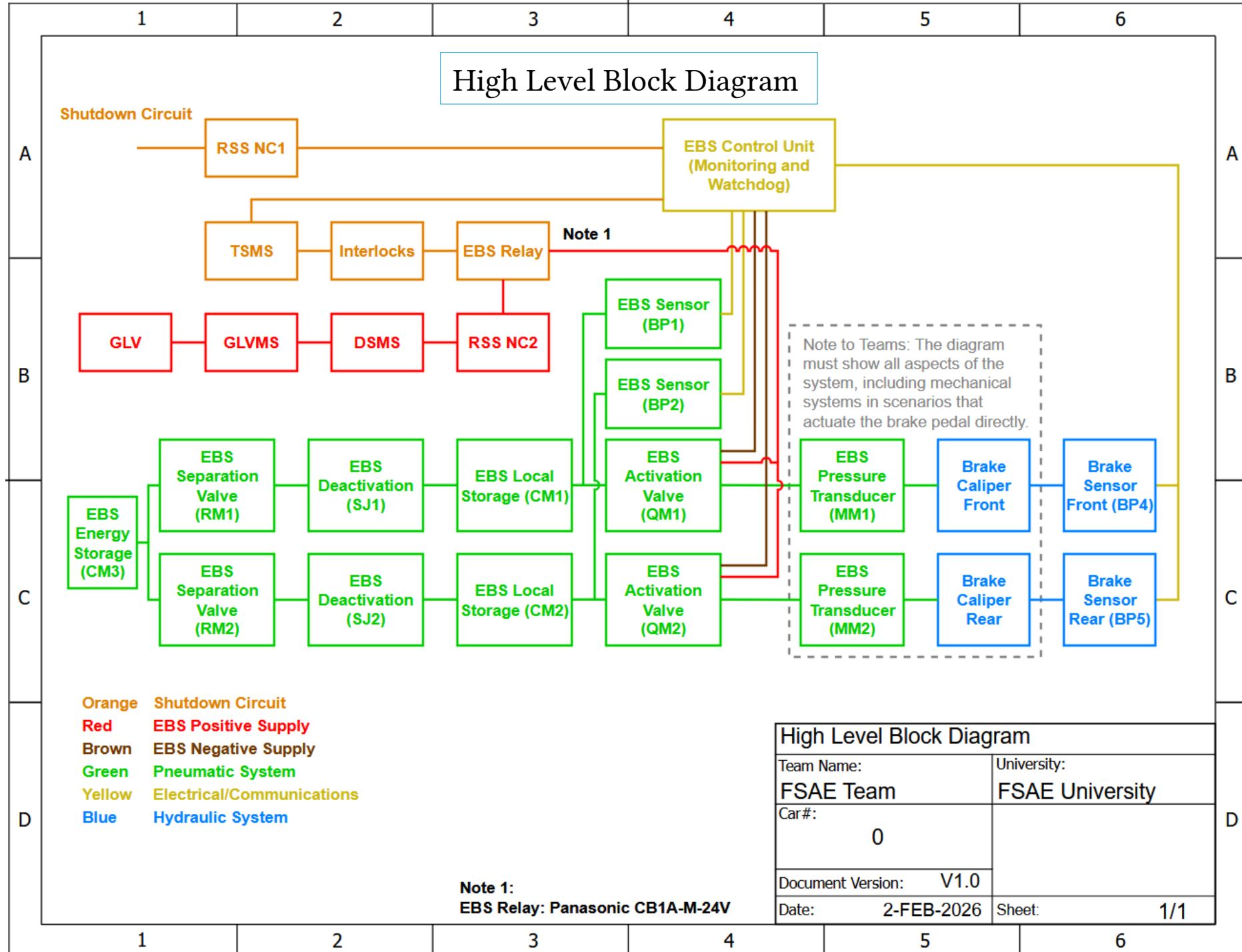


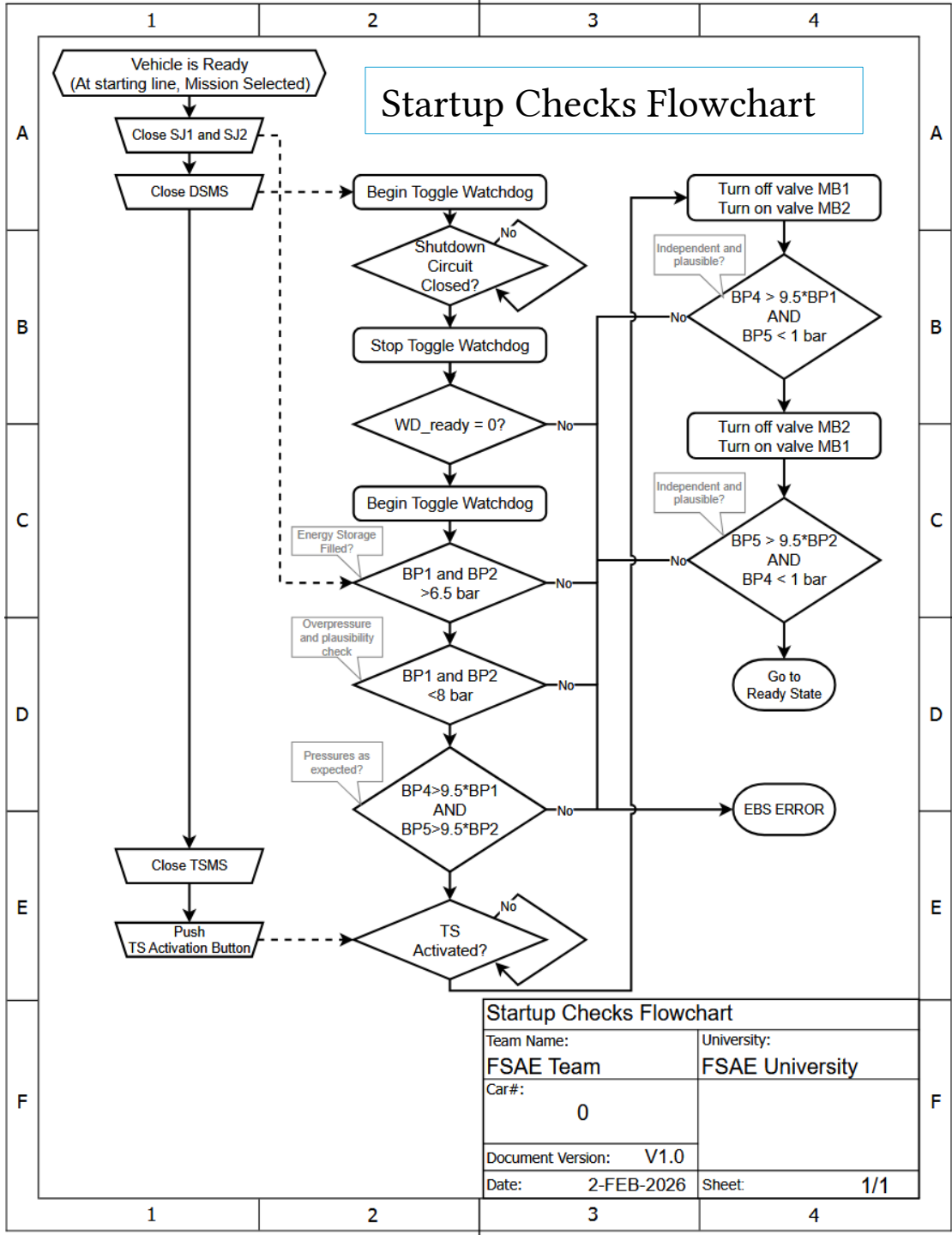
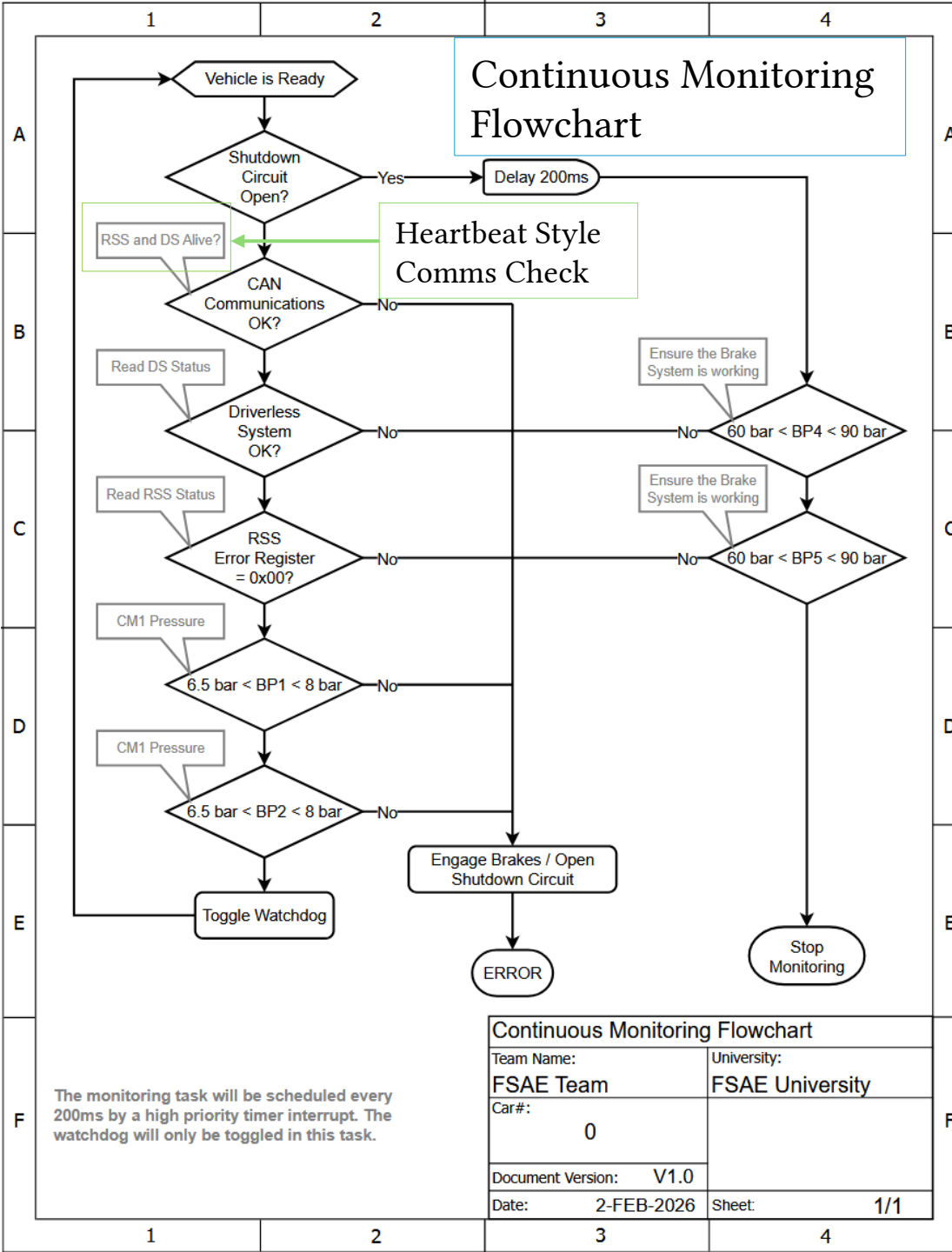
Figure 1. RSS Control Box and Vehicle Module (North American device colors may differ)

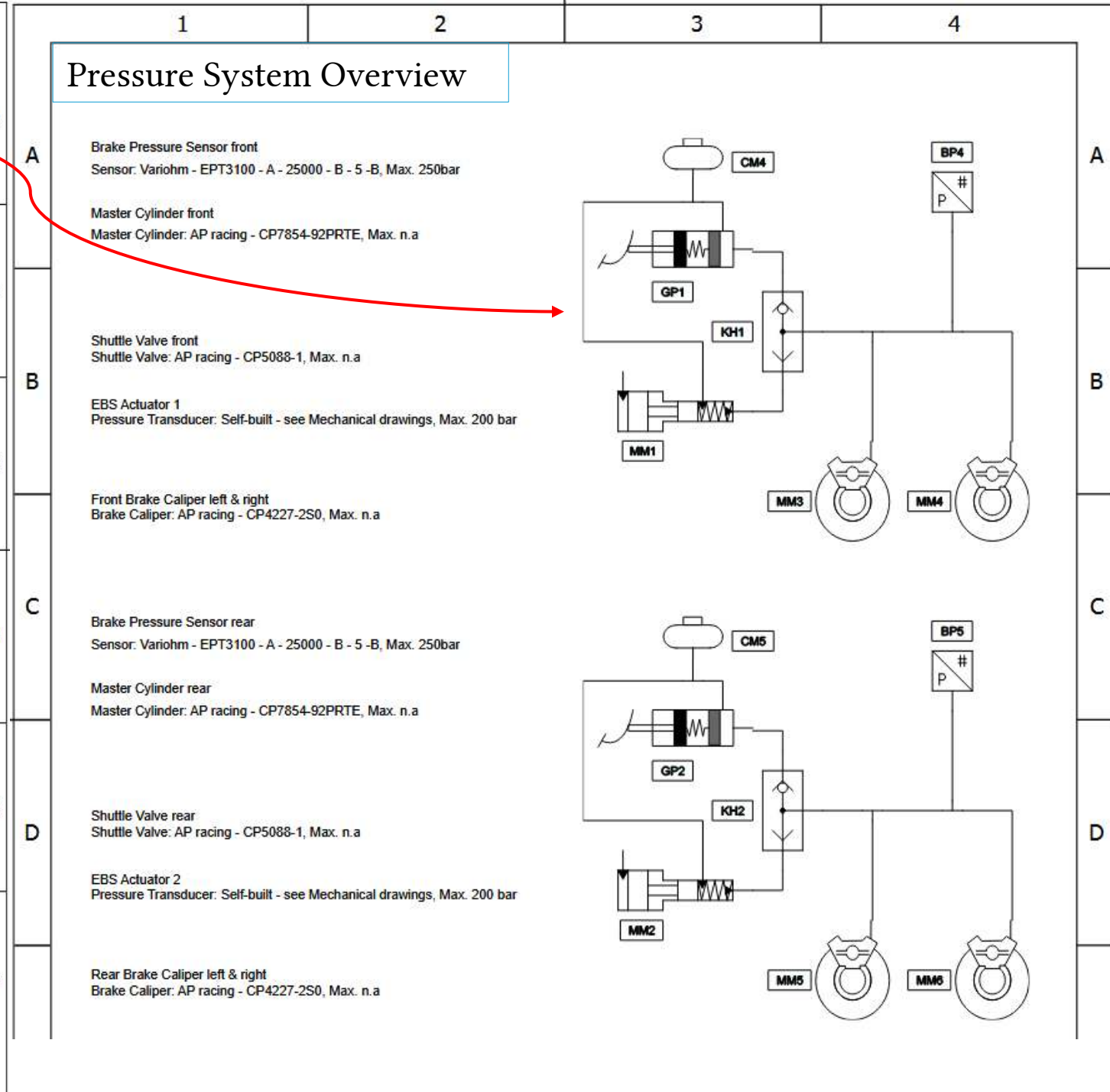
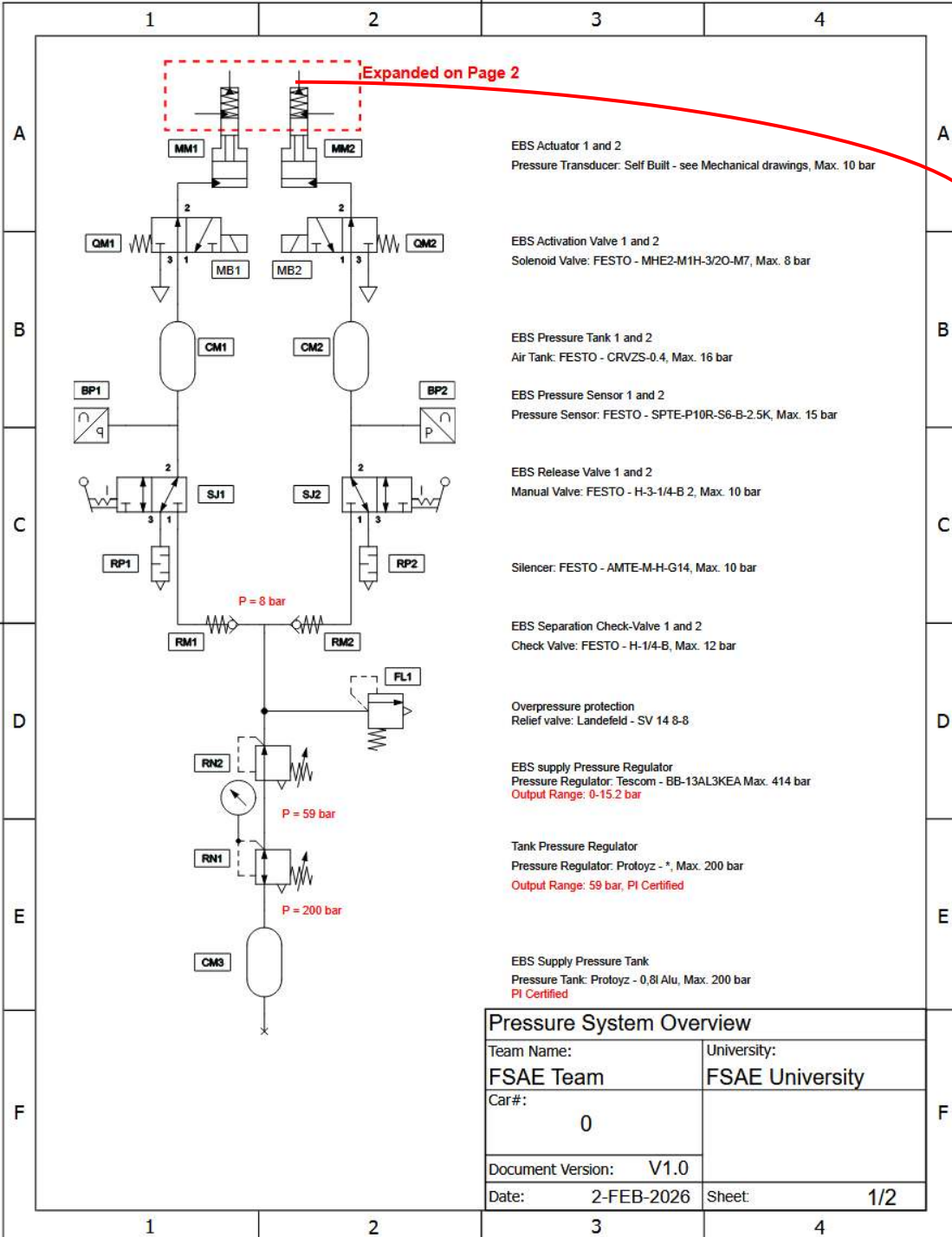
- STOP: Remote Kill Switch opens shut down circuit
- GO: Preselected driverless mission starts

Note: Component locations aren't final, just overlay from last slide





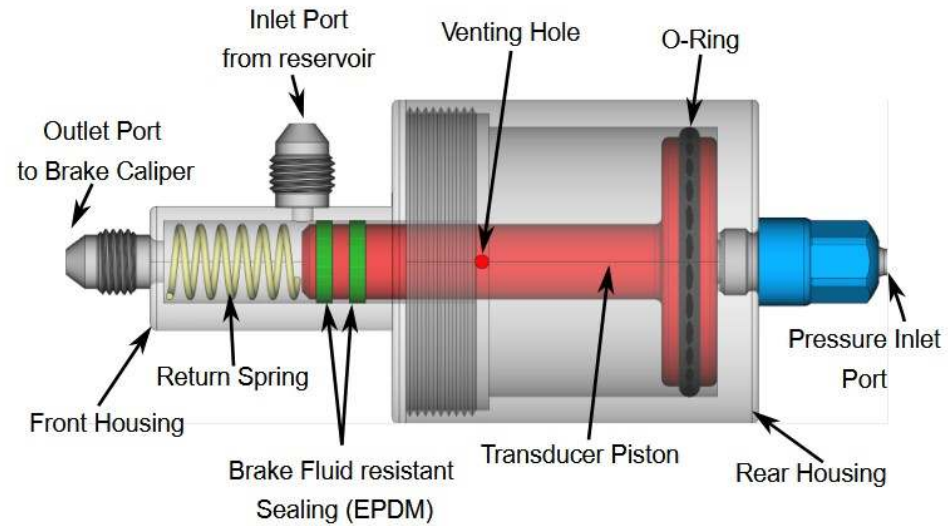




EBS Actuator 1&2 (MM1 / MM2)

$d1 = 38 \text{ mm}$, $d2 = 12 \text{ mm}$, Transfer Ratio = 1:10

Spring preload = 33 N (0.3 bar pneumatic)



Partial Mech Assembly

Mechanical Assembly

Team Name:

FSAE Team

Car#:

0

Document Version:

V1.0

Date:

2-FEB-2026

Sheet:

1/1

University:

FSAE University

NS_:

NS_DESC_
 CM_
 BA_DEF_
 BA_
 VAL_
 CAT_DEF_
 CAT_
 FILTER
 BA_DEF_DEF_
 EV_DATA_
 ENVVAR_DATA_
 SGTYPE_
 SGTYPE_VAL_
 BA_DEF_SGTYPE_
 BA_SGTYPE_
 SIG_TYPE_REF_
 VAL_TABLE_
 SIG_GROUP_
 SIG_VALTYPE_
 SIGTYPE_VALTYPE_
 BO_TX_BU_
 BA_DEF_REL_
 BA_REL_
 BA_DEF_DEF_REL_
 BU_SG_REL_
 BU_EV_REL_
 BU_BO_REL_
 SG_MUL_VAL_

BS_:

BU_: MC IMU

BO_ 67 IMU3_Data: 6 IMU

SG_w_GyroZ1 : 32|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

BO_ 64 IMU0_Data: 8 IMU

SG_G_AccelY1 : 16|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

SG_G_AccelX1 : 0|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

CAN DBC of Driverless Signals

BO_ 66 IMU2_Data: 4 IMU

SG_w_GyroY1 : 16|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

SG_w_GyroX1 : 0|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

BO_ 256 SystemStatus: 5 MC

SG_BrakeLightStatus : 32|1@1- (1,0) [0|0] "" Vector__XXX

SG_GoMission : 12|1@0- (1,0) [0|0] "" Vector__XXX

BO_ 513 BrakeControl: 8 MC

SG_BrakeState : 0|4@1- (1,0) [0|0] "" Vector__XXX

BO_ 65 IMU1_Data: 6 IMU

SG_G_AccelZ1 : 32|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

BA_DEF_SG_ "GenSigSendType" ENUM

"Cyclic","OnWrite","OnWriteWithRepetition","OnChange","OnChangeWithRepetition","IfActive","IfActiveWithRepetition","NoSigSendType";

BA_DEF_SG_ "GenSigInactiveValue" INT 0 0;

BA_DEF_BO_ "GenMsgCycleTime" INT 0 0;

BA_DEF_BO_ "GenMsgSendType" ENUM

"Cyclic","not_used","not_used","not_used","not_used","Cyclic","not_used","IfActive","NoMsgSendType";

BA_DEF_BU_ "NmStationAddress" HEX 0 0;

BA_DEF_ "DBName" STRING ;

BA_DEF_ "BusType" STRING ;

BA_DEF_DEF_ "GenSigSendType" "Cyclic";

BA_DEF_DEF_ "GenSigInactiveValue" 0;

BA_DEF_DEF_ "GenMsgCycleTime" 0;

BA_DEF_DEF_ "GenMsgSendType" "NoMsgSendType";

BA_DEF_DEF_ "NmStationAddress" 0;

BA_DEF_DEF_ "DBName" "";

BA_DEF_DEF_ "BusType" "CAN";

BA_ "BusType" "CAN";

BA_ "DBName" "IMU_Controller_V1";

Resources from FSAE

1. Driverless Roadmap
2. Driverless Rules
3. DSF Outline (and somewhat of a template)
4. RSS Info
5. Driverless Notice of Intent

Resources from FSG

LOTS they've been doing driverless since 2016

- <https://arwo.hamburg/talks/> (videos/lectures)
- FSG website (videos/lectures)
 - Example ASF (very similar to DSF) submissions
 - DSF is a WIP version of the ASF

Rules for FSG 2026

 **FS-Rules_2026_v1.1.pdf** (884 KB)

 **FSG26_Event_Handbook_v1.1.pdf** (687 KB)

Changelog

20251012: FS-Rules 2026 v1.0

20251125: FS Rules 2026 v1.1

20251207: AS Beginners Guide 2026 v1.0 (fsg.one/as-guide)

20251229: FSG26 Event Handbook v1.0

20260104: ASF Examples v26.0 (fsg.one/handson-asf)

20260115: FSG26 Event Handbook v1.0.1

20260123: FSG26 IAD Form v1.0

20260126: FS26 ASes Template v1.0

20260127: ASF Examples v26.1 (fsg.one/handson-asf)

20260127: FSG26 Event Handbook v1.1

Why take this on at all?

- <https://mitadmissions.org/blogs/entry/because-racecar-friendships-and-family/>





DRIVERLESS

**COME ON DO
SOMETHING...**

